

HAZOP REPORT

Vacuum Batch Distillation Ver 1 (pilot column C-101)

Methyl palmitate (C16:0) from palm-oil biodiesel · pilot vacuum batch column C-101

Document no.	Revision	Status
VBD-SA-HZP-001	A	Issued for teaching

Revision record

Rev	Date	Description	Prepared	Checked	Approved
A		Issued for teaching (HAZOP study of the OTS model)	HAZOP scribe	HAZOP leader	Project lead

This is a teaching HAZOP of an operator training simulator model. Names, signatures and dates are left blank for the real study team. Numbers come from the simulator model (alarm limits, trip settings, runs) unless marked as assumed or as from the project report.

HAZOP CLOSE-OUT SUMMARY

33 actions from 76 worksheet rows. Status is OPEN for every action at rev A; the action-taken column is filled by the action party.

Node	Action no.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N3 N5 N7 N8	R01	N3.08 N5.07 N7.03 N8.16	Leak-test C-101 and the vacuum line before every heat-up (pressure-rise test with VP-101 isolated) and after any flange is opened; record the rise rate against an acceptance limit.	Operations / Process		OPEN	Highest risk addressed: 12 High
N3 N6 N7 N8	R02	N3.08 N6.09 N7.03 N7.09 N8.11	Add an O ₂ analyser or a leak-rate estimate (from the PV-102 output) on the vacuum line, alarmed. Use LOPA to decide whether an air-ingress SIF (trip heating and purge with N ₂) is needed: a 12× leak does not reach PSHH-102.	Instrument / Safety		OPEN	Highest risk addressed: 12 High
N7 N8	R03	N7.09 N8.10 N8.11 N8.12	Interlock HV-110: open only when the pot (TE-101B) is below 150 °C, N ₂ supply only (dedicated coupling, no air connection), N ₂ low-pressure alarm.	Instrument / Process		OPEN	Highest risk addressed: 12 High
N3 N7	R04	N3.08 N7.02 N7.03	Alarm the PV-102 output when it falls below ≈ 20 % (assumed): little spare vacuum capacity left, an early sign of a leak or of VP-101 wear.	Control		OPEN	Highest risk addressed: 12 High
N4 N7 N8	R05	N4.01 N7.01 N7.08 N8.12	Fit a non-return and an automatic isolation valve on the VP-101 suction; add a VP-101 run-status alarm.	Mechanical / Instrument		OPEN	Highest risk addressed: 9 Medium
N1 N4 N5 N7	R06	N1.06 N4.01 N4.07 N5.01 N7.02 N7.06 N7.07	Route the VP-101 exhaust outdoors through a demister; check and empty the trap every batch; consider a cold trap.	Process		OPEN	Highest risk addressed: 9 Medium
N7	R07	N7.04 N7.05	Review PV-102: when it fails open the top settles at about the PSHH-102 setting. Limit PV-102 travel or move the trip setting so that the outcome is defined.	Process / Instrument		OPEN	Highest risk addressed: 6 Medium
N2	R08	N2.04	Add a TT-107 vs TE-107B deviation alarm (e.g. 10 °C, assumed) so that a drifting TT-107 is seen before TSHH-107 trips.	Instrument		OPEN	Highest risk addressed: 8 Medium
N1 N2	R09	N1.10 N2.01 N2.06 N2.07 N2.08 N2.09	Confirm the hot-oil loop protection: relief and expansion tank, P-101 minimum-flow bypass, hard-wired HE-101 no-flow interlock (flow switch), expansion-tank low-level alarm, oil analysis each year.	Process / Mechanical		OPEN	Highest risk addressed: 8 Medium
N1 N2	R10	N1.09 N2.07	Fit spray shields on hot-oil flanges and the P-101 seal; inspect for leaks before each campaign; extinguisher at HE-101.	Safety / Mechanical		OPEN	Highest risk addressed: 8 Medium
N2	R11	N2.05	Alarm when HV-104 (E-104 cooler) is open while heating is armed.	Control		OPEN	Highest risk addressed: 3 Low
N1 N2 N3 N4 N7 N8	R12	N1.05 N1.07 N2.02 N2.03 N3.01 N3.04 N4.07 N7.04 N8.04	Clamp the JIC-101 set point at 5 kW once the pot boils and add a PDI-103 high-high override in the BPCS that cuts duty; train the heat-up duty cut.	Control / Operations		OPEN	Highest risk addressed: 8 Medium
N1	R13	N1.02	Give LSL-101 its own level transmitter (it shares LT-101 with LI-101); confirm the SIL by LOPA.	Instrument / Safety		OPEN	Highest risk addressed: 9 Medium
N1 N3 N6 N8	R14	N1.02 N1.03 N3.05 N6.03 N8.08 N8.09	Write the end-of-campaign rule as a stop on the first of: lab result, TI-103 rise, pot level 35 % (assumed); then stop heating at once. Add a TI-103 rate-of-rise alarm.	Operations / Process		OPEN	Highest risk addressed: 9 Medium
N1	R15	N1.01	Add a high-level alarm on LI-101 for charging and charge through a meter or by weight.	Instrument / Operations		OPEN	Highest risk addressed: 6 Medium
N1 N4 N5 N7 N8	R16	N1.06 N4.05 N5.08 N7.07 N8.01	Analyse every charge before charging (water, methanol, C16:0) and adapt the cut plan to the feed.	Operations / Laboratory		OPEN	Highest risk addressed: 6 Medium

Node	Action no.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N1 N3 N8	R17	N1.07 N3.04 N8.14	Drain the residue fully every batch, inspect D-102, and keep an antifoam procedure.	Operations		OPEN	Highest risk addressed: 6 Medium
N1	R18	N1.03 N1.08	Track the reboiler UA per batch (duty vs TV-101 output and TI-101 – TI-106); clean D-102 when it falls below 70 % of new (assumed).	Process		OPEN	Highest risk addressed: 9 Medium
N4	R19	N4.03	Add a TI-109 low-low alarm and make the tempered-water TCV fail towards warm; consider stopping heating when TI-109 stays below 40 °C (assumed).	Instrument / Process		OPEN	Highest risk addressed: 9 Medium
N4	R20	N4.02	Fit a duplex strainer with a ΔP gauge upstream of HV-105.	Mechanical		OPEN	Highest risk addressed: 9 Medium
N4	R21	N4.03 N4.06	Warm E-103 to 60 °C before the first vapour arrives and never leave the tempered water cold after shutdown.	Operations		OPEN	Highest risk addressed: 9 Medium
N3 N5 N7	R22	N3.03 N5.01 N5.02 N7.06	Add LSHH-104 (D-104 high-high) or use the E-103 overload signal to cut heating; LOPA to size it. A full drum sends ester to VP-101 with no trip.	Instrument / Safety		OPEN	Highest risk addressed: 9 Medium
N3 N5 N8	R23	N3.05 N5.03 N5.05 N8.07	Limit RY-104 entries (off-cut 10-25, product 8-20, assumed) and add R and HS-107 to the shift handover check.	Control / Operations		OPEN	Highest risk addressed: 8 Medium
N5 N6 N8	R24	N5.05 N6.01 N8.07	Fit XV-107 position switches with a discrepancy alarm against HS-107, and a WI-201 / WI-202 rate check.	Instrument		OPEN	Highest risk addressed: 8 Medium
N3 N6 N8	R25	N3.06 N6.02 N6.08 N6.09 N8.06	Switch cuts on the D-105 lab result, not on TI-104 alone; flush the sample line; use a closed sample pot isolated from the vacuum.	Operations / Process		OPEN	Highest risk addressed: 8 Medium
N5 N6 N8	R26	N5.07 N6.07 N8.13	Add a sample cooler at D-105 and a product cooler before TK-201 / TK-202; face shield and gloves for sampling and draining.	Safety / Process		OPEN	Highest risk addressed: 9 Medium
N6	R27	N6.06	Add WI-201 / WI-202 high-weight alarms and size the receivers for the largest cut (Table 9 feed).	Process / Instrument		OPEN	Highest risk addressed: 6 Medium
N6	R28	N6.05	Keep the receivers balanced to the vacuum system with a non-return valve; put XV-107 to CLOSED before any receiver is changed.	Process / Operations		OPEN	Highest risk addressed: 4 Low
N1 N8	R29	N1.09 N8.02 N8.13	Drain the residue only below 80 °C (assumed) through a closed drain; double-block the pot drain; do not charge a hot pot.	Operations		OPEN	Highest risk addressed: 9 Medium
N3 N8	R30	N3.07 N8.15	Before opening C-101: cool below 50 °C, purge with N ₂ and keep the packing wet; permit to work.	Safety / Operations		OPEN	Highest risk addressed: 8 Medium
N1 N2 N3 N4 N7 N8	R31	N1.02 N1.03 N2.04 N3.08 N4.01 N4.02 N7.03 N7.06 N8.09 N8.11	Carry every High row and every row that relies on a SIS-100 trip into LOPA; record the SIL targets in the SRS.	Safety		OPEN	Highest risk addressed: 12 High
N6 N8	R32	N6.02 N6.03 N8.08	Alarm the operator at the cut switch and the end point (lab request reminders); staff night shifts of long campaigns.	Operations		OPEN	Highest risk addressed: 8 Medium
N1 N2 N3 N4 N5 N7 N8	R33	N1.03 N1.05 N1.07 N2.01 N2.02 N2.04 N3.01 N3.03 N3.04 N3.08 N4.01 N4.03 N4.07 N5.01 N7.01 N7.03 N7.04 N7.05 N7.06 N8.11	Train the deviations of the dynamic check on the simulator (one per shift team) and revalidate this HAZOP every 5 years or at any MOC.	Operations / Safety		OPEN	Highest risk addressed: 12 High

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Vacuum Batch Distillation Ver 1 (pilot column C-101)

Document VBD-SA-HZP-001 · revision A · Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University

Approvals

Role	Name	Signature	Date
Prepared by: HAZOP scribe			
Checked by: HAZOP leader (independent)			
Reviewed by: process-safety engineer			
Approved by: project lead / pilot-plant manager			

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Executive summary

The study examined 8 nodes of the pilot vacuum batch column — seven equipment nodes and one procedure node (N8) in which the guide words were asked of every batch phase — and recorded 76 deviations with full recording. After the existing safeguards the risk ranks are: 14 Low, 59 Medium, 3 High and 0 Very high. 33 recommendations were raised.

Key findings

1. Air entering the hot column (leak, or vacuum broken while the pot is above 150 °C) is the only High-ranked scenario family (N3.08, N7.03, N8.11). The dynamic check shows that a 12× air leak does not reach the PSHH-102 trip (> 300 mbar): only alarms and the operator stand between the leak and a fire. LOPA is needed for an air-ingress function (R02) and for an HV-110 permissive (R03).
2. The five SIS-100 trips act as intended in the runs: FSLL-108 on loss of tempered water, PSHH-102 on loss of vacuum, TSHH-107 on a hot-oil thermocouple reading low. Each trip removes the heat but leaves the hot inventory; recovery depends on the procedures.
3. Several credible deviations have no trip at all: a full reflux drum (P-104 trip) sends ester to the vacuum trap, TV-101 wide open or PV-102 closed floods the column, and cold tempered water plugs E-103. These are Medium-ranked; the register adds BPCS limits and alarms (R12, R19, R22).
4. PV-102 failing open settles the top pressure at the PSHH-102 setting, so whether the plant trips is undefined (R07). Shared transmitters (LSLL-101 on LT-101, PSHH-102 on PT-102, FSLL-108 on FT-108) limit the credit LOPA can give (R13).

Actions by party

Action party	Actions (lead or shared)
Operations	14
Process	12
Instrument	11
Safety	8
Control	4
Mechanical	4
Laboratory	1

1 Introduction

The unit is a pilot vacuum batch distillation column C-101 that recovers methyl palmitate (C16:0, ≥ 99 wt %) from palm-oil biodiesel. A charge of 110 L is boiled in pot D-101 by a hot-oil reboiler D-102; the column has structured packing; overhead vapour is condensed in E-103 on tempered water and returned from reflux drum D-104 by P-104. A campaign runs at total reflux, then an off-cut to TK-201 at R 15, then the product cut to TK-202 at R 10, and stops before the C18 front; the residue stays in the pot. Vacuum is held by VP-101 with an N₂ bleed (PV-102). SIS-100 protects the heating.

The study was made on the operator training simulator of the unit; the simulator model provides the design conditions, alarm limits, trip settings and the dynamic check. It follows IEC 61882:2016 and is a teaching study: the 5 × 5 risk matrix is an example matrix shared with the HAZID, Safety layers and LOPA work.

2 Scope

Boundaries. From the charge in D-101 to the receivers TK-201 / TK-202; the hot-oil loop from HE-101 to D-102 and back, including E-104; the tempered-water supply at E-103; the vacuum system to the VP-101 exhaust. Outside: biodiesel production and feed drying, the internal controls of the hot-oil heater and the tempered-water heater, utilities and the building.

2.1 Nodes

Node	Description	Design intent	Design conditions	Rows
N1	Pot D-101 and reboiler D-102 (process side)	Hold the 110 L charge (95.7 kg) in D-101 and boil it at pot pressure 171 mbar with 3.0 kW from D-102, sending the vapour into C-101; the residue (C18 + C20 rich) stays in the pot at the end.	Pot 255.5 °C, 171 mbar, level 89 % at the start of the campaign	10
N2	Hot-oil loop (HE-101, P-101, XV-101, TV-101, E-104)	Circulate hot oil at 290 °C (TIC-107 moves HE-101, 12 kW) with P-101 through XV-101 and TV-101 to D-102 so that the pot receives the duty set on JIC-101 (3.0 kW at design, TV-101 ≈ 40 %); E-104 / HV-104 cools the oil only at cool-down.	Supply 290 °C, return 286 °C, duty 3.0 kW	9
N3	Column C-101 (packing, profile, ΔP)	Separate the vapour from the pot over 20 equilibrium stages (structured packing, Murphree E 0.6) at 160 mbar top, ΔP 11 mbar and flooding index 61 % so that the light esters and then C16:0 leave at the top.	TI-104 192.4 °C, TI-103 193.2 °C, TI-102 237 °C at total reflux; ΔP 10.9 mbar	8
N4	Condenser E-103 and tempered water (P-108, HV-105)	Condense all overhead vapour at 160 mbar in E-103 with tempered water at 60 °C and 1.2 m ³ /h; keep the water above 45 °C so that C16/C18 esters (m.p. 30–38 °C) do not freeze on the tubes.	Water 60 °C, 1.20 m ³ /h; condensate TI-105 176 °C at total reflux	7
N5	Reflux drum D-104, P-104, FV-104 / FV-105	Hold D-104 (4 L) at 50 % with LIC-104 and return reflux through FV-104 (fail open) at the ratio set on RY-104 (R 15 off-cut, R 10 product); the distillate leaves through FV-105 (fail closed).	Drum 51 %, reflux 32 kg/h at total reflux; product cut: reflux 34.9 kg/h, distillate 3.49 kg/h	8
N6	Distillate routing XV-107 / HS-107, D-105, TK-201, TK-202	Send no distillate at total reflux, the off-cut (C8–C14) to TK-201 at R 15 and the product (C16:0 ≥ 99 wt %) to TK-202 at R 10, switching on the D-105 sample; stop the product cut before the C18 front.	Reference campaign: switch at 307 min, end at 907 min, TK-201 11.1 kg, TK-202 34.7 kg at 99.0 % C16:0	9
N7	Vacuum system (VP-101, trap, PV-102 N ₂ bleed, HV-110)	Hold the column top at 160 mbar with VP-101 and the PV-102 N ₂ bleed (fail open, ≈ 50 % at design) against an air in-leakage of 20 (model units); collect uncondensed ester in the trap; HV-110 breaks vacuum with N ₂ only when the pot is below 150 °C.	Top 160 mbar, pot 171 mbar; PSHH-102 active only while heating is armed	9
N8	Procedures (batch phases)	Run the batch safely through each phase: charging, heat-up, total reflux, off-cut, product cut, end and cool-down, vacuum break, discharge of the residue, and maintenance.	Reference campaign A: off-cut 307 min, end 907 min; cool-down to < 150 °C before the vacuum break	16

2.2 Node map

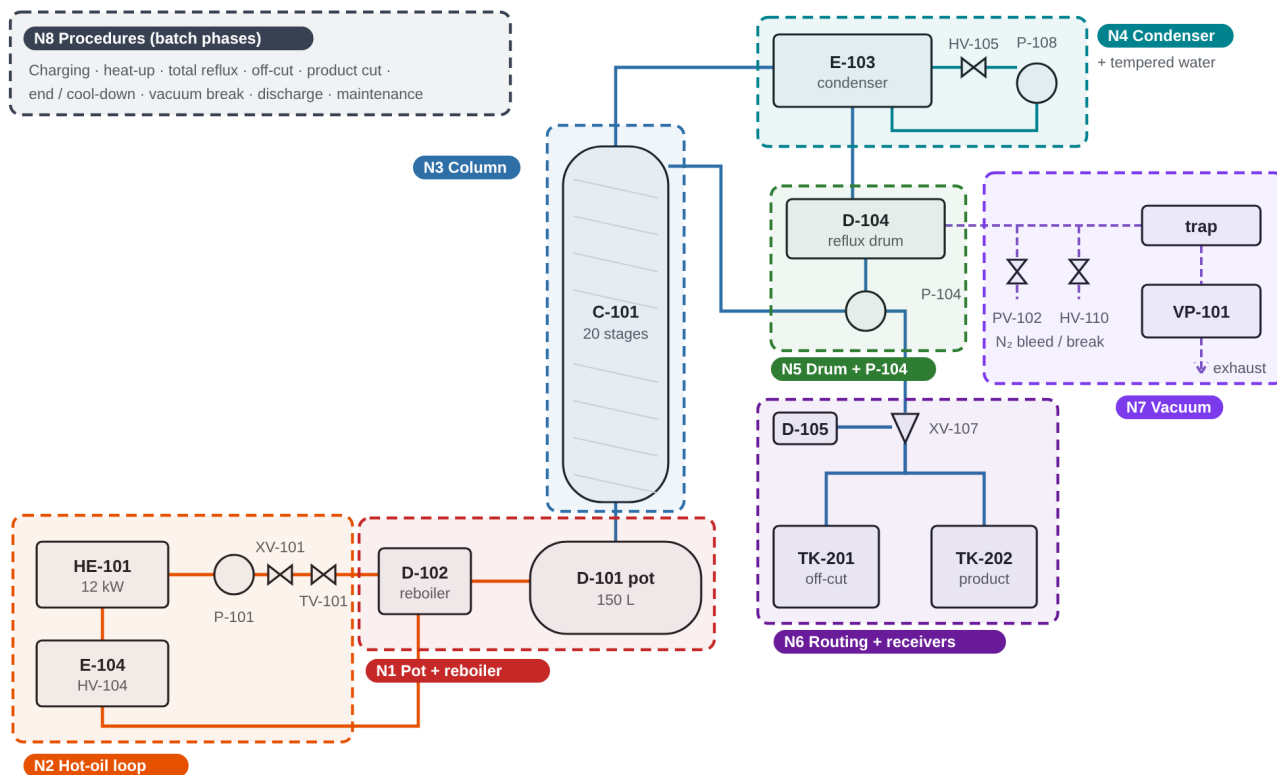


Figure 1 · Node map (coloured dashed boxes over simplified equipment blocks; N8 = procedures).

3 Methodology

The study follows IEC 61882:2016 in four phases: definition (scope, objectives, team), preparation (documents, nodes, design intents), examination and documentation with follow-up. For every node the team states the design intent, then combines each parameter with each guide word; only physically meaningful deviations are recorded. Consequences are judged with no credit for safeguards; the likelihood is judged with the existing safeguards. Full recording is used.

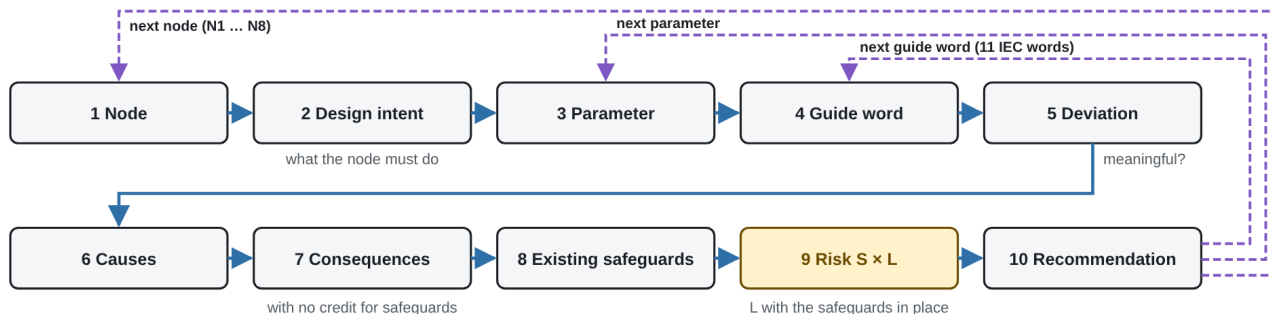


Figure 2 · The examination loop.

3.1 Guide words

Guide word	Meaning	Example from this study
NO / NOT	Complete negation of the design intent	N2.01 Flow: No hot-oil flow to D-102
MORE	Quantitative increase	N1.01 Level: Pot overfilled at charging (above the 110 L charge)
LESS	Quantitative decrease	N1.02 Level: Pot boiled down at the end of the campaign
AS WELL AS	Qualitative increase: something extra is present	N1.07 Composition: Foam in the pot (surface-active degradation products)
PART OF	Qualitative decrease: only part of the intent is achieved	N2.03 Flow: Duty arrives in pulses (as-built on/off heating)
REVERSE	Logical opposite of the intent (reverse flow)	N1.10 Flow: Hot oil leaks into the process side (D-102 tube leak)
OTHER THAN	Complete substitution: something else happens	N1.06 Composition: Wrong or wet feed charged (water, methanol traces, other FAME)
EARLY	Relative to clock time: happens too early	N6.02 Routing: Switch to product too early
LATE	Relative to clock time: happens too late	N6.03 Routing: Product cut stopped too late
BEFORE	Relative to order or sequence: happens before it should	N6.04 Routing: Distillate drawn before total reflux has lined out
AFTER	Relative to order or sequence: happens after it should	N8.06 Off-cut: Decision taken on a lab result that arrives after the change it should guide

3.2 Parameters

Parameter	In this unit
Flow	vapour boil-up, reflux, distillate, hot oil, tempered water, N ₂ bleed, vapour to VP-101
Pressure	top (PIC-102), pot (PI-103), column ΔP (PDI-103), hot-oil loop
Temperature	pot, packing profile, hot oil, tempered water, condensate
Level	pot D-101, reflux drum D-104, receivers TK-201/TK-202
Composition	charge, distillate cut, air/O ₂ in the hot system, water or methanol traces
Heat duty	pot duty JIC-101, condenser load
Reflux ratio	RY-104 split (R 15 off-cut, R 10 product)
Routing	HS-107 / XV-107 destination
Phase	freezing of C16/C18 esters, foam
Containment	leaks of hot ester or hot oil, air in-leakage
Sequence / timing	batch phases of node N8 (charging ... discharge)

3.3 Risk matrix

$R = S \times L$. Bands: 1-4 Low, 5-9 Medium, 10-16 High, 20-25 Very high. The cells give the number of rows of this study.

S \ L	1 Remote	2 Unlikely	3 Possible	4 Likely	5 Frequent
5 Catastrophic	.	.	.	.	.
4 Major	1	10	3	.	.
3 Serious	1	6	9	.	.
2 Moderate	.	5	23	11	.
1 Minor	.	1	6	.	.

S	Severity	Description
1	Minor	First aid; product rework only; no release
2	Moderate	Batch off-spec or lost; minor equipment damage; small contained spill
3	Serious	Lost-time injury (burn); equipment damage; release of ester vapour indoors
4	Major	Fire in the unit; permanent injury; major equipment damage
5	Catastrophic	Fatality; fire spreading beyond the pilot hall

L	Likelihood	Frequency band (with existing safeguards)
1	Remote	Not expected in the life of the pilot ($< 10^{-3}$ /yr)
2	Unlikely	Could happen once in the life of the pilot (10^{-3} – 10^{-2} /yr)
3	Possible	Once in a few years (10^{-2} – 10^{-1} /yr)
4	Likely	About once a year or a few times a year (0.1–1 /yr)
5	Frequent	Several times per year or more (> 1 /yr)

Risk	R	Action
Low	1–4	Acceptable; manage by procedure and normal supervision
Medium	5–9	Tolerable if ALARP; recommendations to be considered and closed
High	10–16	Not acceptable long term; risk reduction required; LOPA to size the protection
Very high	20–25	Not acceptable; do not operate until reduced

4 Sessions

4.1 Schedule

Session	Date	Nodes	Duration
1		Introduction, N1, N2	
2		N3, N4	
3		N5, N6	
4		N7, N8 (procedures)	
5		Review of the register and the dynamic check	

4.2 Team

Role	Name	Brings
HAZOP leader (independent)		Method, timing, guide-word discipline
Scribe		Full recording; reads back recommendations
Process engineer		Design intent, balances, operating windows
Operations (senior operator)		Charging, cut switching, cool-down as really done
Instrument / control engineer		Loops, fail positions, SIS-100 logic and settings
Process-safety engineer		Risk matrix, links to LOPA and the SRS
Specialists as needed		Hot-oil and vacuum equipment, laboratory (GC)

4.3 Documents reviewed

Document	Used for
Screen P&ID of the OTS (DWG VBD-V1-PID-001 rev A, Appendix 2)	Tags, fail positions, safeguards; no EPC P&ID was drawn for this study
Process and Design tabs of the OTS	Design intent, balances, sizes, reference campaign
Control loop design tab	Loops and controller modes
SIS-100 and campaign tab	Trip settings and delays 5, 5, 10, 5, 5 s, arming permissives
DCS console alarm list	11 alarm limits
HAZID tab	Hazard register and anchor risk rankings
C&E matrix and SRS of SIF tabs	Trip effects and SIF requirements (read alongside this report)
Operating procedures of the OTS	Start-up, campaign A and B, shutdown, loss of cooling

4.4 Guide-word coverage

Node	NO	MORE	LESS	AS WELL AS	PART OF	REVERSE	OTHER THAN	EARLY	LATE	BEFORE	AFTER	Rows
N1	·	3	3	1	·	1	2	·	·	·	·	10
N2	1	3	1	1	1	1	1	·	·	·	·	9
N3	1	3	2	1	1	·	·	·	·	·	·	8
N4	1	2	2	1	·	1	·	·	·	·	·	7
N5	1	3	2	1	·	·	1	·	·	·	·	8
N6	·	2	·	1	1	1	1	1	1	1	·	9
N7	·	4	1	1	·	1	2	·	·	·	·	9
N8	1	1	1	1	2	·	2	2	1	3	2	16
All	5	21	12	8	5	5	9	3	2	4	2	76

All 11 of 11 guide words are used; a dot means the combination was considered and gave nothing meaningful.

5 Findings

5.1 Risk profile per node

Node	Rows	Low	Medium	High	Very high	Highest
N1 Pot D-101 and reboiler D-102 (process side)	10	2	8	0	0	9
N2 Hot-oil loop (HE-101, P-101, XV-101, TV-101, E-104)	9	3	6	0	0	8
N3 Column C-101 (packing, profile, ΔP)	8	1	6	1	0	12
N4 Condenser E-103 and tempered water (P-108, HV-105)	7	2	5	0	0	9
N5 Reflux drum D-104, P-104, FV-104 / FV-105	8	3	5	0	0	9
N6 Distillate routing XV-107 / HS-107, D-105, TK-201, TK-202	9	2	7	0	0	9
N7 Vacuum system (VP-101, trap, PV-102 N ₂ bleed, HV-110)	9	0	8	1	0	12
N8 Procedures (batch phases)	16	1	14	1	0	12
All	76	14	59	3	0	12

5.2 High-ranked rows and rows for LOPA

Every High row, and every row with $S \geq 3$ that relies on a SIS-100 trip, goes to LOPA (R31).

Row	Deviation	S	L	Risk	SIS-100 function relied on	Rec.
N1.02	Level / LESS: Pot boiled down at the end of the campaign	3	3	9 Medium	SIF-01 TSHH-101, SIF-04 LSL-101	R13 R14 R31
N1.03	Temperature / MORE: Pot temperature high (above the TIC-101 alarm)	3	3	9 Medium	SIF-01 TSHH-101	R14 R18 R31 R33
N2.04	Temperature / MORE: Hot oil above 290 °C	4	2	8 Medium	SIF-05 TSHH-107	R08 R31 R33
N3.08	Composition / AS WELL AS: Air enters the hot column (in-leakage)	4	3	12 High	SIF-02 PSHH-102	R01 R02 R04 R31 R33
N4.01	Flow / NO: No tempered water to E-103	3	3	9 Medium	SIF-02 PSHH-102, SIF-03 FSLL-108	R05 R06 R31 R33
N4.02	Flow / LESS: Tempered-water flow low	3	3	9 Medium	SIF-03 FSLL-108	R20 R31
N7.03	Composition / AS WELL AS: Air in-leakage into the hot system	4	3	12 High	SIF-02 PSHH-102	R01 R02 R04 R31 R33
N7.06	Flow / MORE: Uncondensed ester to VP-101	3	2	6 Medium	SIF-03 FSLL-108	R06 R22 R31 R33
N8.09	End / cool-down / LATE: Heating not stopped at the end of the campaign	3	3	9 Medium	SIF-01 TSHH-101, SIF-04 LSL-101	R14 R31
N8.11	Vacuum break / EARLY: Vacuum broken while the pot is above 150 °C	4	3	12 High	SIF-02 PSHH-102	R03 R02 R31 R33

5.3 Recommendation register

ID	Recommendation	Node	From rows	Highest risk	Action by
R01	Leak-test C-101 and the vacuum line before every heat-up (pressure-rise test with VP-101 isolated) and after any flange is opened; record the rise rate against an acceptance limit.	N3 N5 N7 N8	N3.08 N5.07 N7.03 N8.16	12 High	Operations / Process
R02	Add an O ₂ analyser or a leak-rate estimate (from the PV-102 output) on the vacuum line, alarmed. Use LOPA to decide whether an air-ingress SIF (trip heating and purge with N ₂) is needed: a 12× leak does not reach PSHH-102.	N3 N6 N7 N8	N3.08 N6.09 N7.03 N7.09 N8.11	12 High	Instrument / Safety
R03	Interlock HV-110: open only when the pot (TE-101B) is below 150 °C, N ₂ supply only (dedicated coupling, no air connection), N ₂ low-pressure alarm.	N7 N8	N7.09 N8.10 N8.11 N8.12	12 High	Instrument / Process
R04	Alarm the PV-102 output when it falls below ≈ 20 % (assumed): little spare vacuum capacity left, an early sign of a leak or of VP-101 wear.	N3 N7	N3.08 N7.02 N7.03	12 High	Control
R05	Fit a non-return and an automatic isolation valve on the VP-101 suction; add a VP-101 run-status alarm.	N4 N7 N8	N4.01 N7.01 N7.08 N8.12	9 Medium	Mechanical / Instrument
R06	Route the VP-101 exhaust outdoors through a demister; check and empty the trap every batch; consider a cold trap.	N1 N4 N5 N7	N1.06 N4.01 N4.07 N5.01 N7.02 N7.06 N7.07	9 Medium	Process
R07	Review PV-102: when it fails open the top settles at about the PSHH-102 setting. Limit PV-102 travel or move the trip setting so that the outcome is defined.	N7	N7.04 N7.05	6 Medium	Process / Instrument
R08	Add a TT-107 vs TE-107B deviation alarm (e.g. 10 °C, assumed) so that a drifting TT-107 is seen before TSHH-107 trips.	N2	N2.04	8 Medium	Instrument

ID	Recommendation	Node	From rows	Highest risk	Action by
R09	Confirm the hot-oil loop protection: relief and expansion tank, P-101 minimum-flow bypass, hard-wired HE-101 no-flow interlock (flow switch), expansion-tank low-level alarm, oil analysis each year.	N1 N2	N1.10 N2.01 N2.06 N2.07 N2.08 N2.09	8 Medium	Process / Mechanical
R10	Fit spray shields on hot-oil flanges and the P-101 seal; inspect for leaks before each campaign; extinguisher at HE-101.	N1 N2	N1.09 N2.07	8 Medium	Safety / Mechanical
R11	Alarm when HV-104 (E-104 cooler) is open while heating is armed.	N2	N2.05	3 Low	Control
R12	Clamp the JIC-101 set point at 5 kW once the pot boils and add a PDI-103 high-high override in the BPCS that cuts duty; train the heat-up duty cut.	N1 N2 N3 N4 N7 N8	N1.05 N1.07 N2.02 N2.03 N3.01 N3.04 N4.07 N7.04 N8.04	8 Medium	Control / Operations
R13	Give LSL-101 its own level transmitter (it shares LT-101 with LI-101); confirm the SIL by LOPA.	N1	N1.02	9 Medium	Instrument / Safety
R14	Write the end-of-campaign rule as a stop on the first of: lab result, TI-103 rise, pot level 35 % (assumed); then stop heating at once. Add a TI-103 rate-of-rise alarm.	N1 N3 N6 N8	N1.02 N1.03 N3.05 N6.03 N8.08 N8.09	9 Medium	Operations / Process
R15	Add a high-level alarm on LI-101 for charging and charge through a meter or by weight.	N1	N1.01	6 Medium	Instrument / Operations
R16	Analyse every charge before charging (water, methanol, C16:0) and adapt the cut plan to the feed.	N1 N4 N5 N7 N8	N1.06 N4.05 N5.08 N7.07 N8.01	6 Medium	Operations / Laboratory
R17	Drain the residue fully every batch, inspect D-102, and keep an antifoam procedure.	N1 N3 N8	N1.07 N3.04 N8.14	6 Medium	Operations
R18	Track the reboiler UA per batch (duty vs TV-101 output and TI-101 – TI-106); clean D-102 when it falls below 70 % of new (assumed).	N1	N1.03 N1.08	9 Medium	Process
R19	Add a TI-109 low-low alarm and make the tempered-water TCV fail towards warm; consider stopping heating when TI-109 stays below 40 °C (assumed).	N4	N4.03	9 Medium	Instrument / Process
R20	Fit a duplex strainer with a ΔP gauge upstream of HV-105.	N4	N4.02	9 Medium	Mechanical
R21	Warm E-103 to 60 °C before the first vapour arrives and never leave the tempered water cold after shutdown.	N4	N4.03 N4.06	9 Medium	Operations
R22	Add LSHH-104 (D-104 high-high) or use the E-103 overload signal to cut heating; LOPA to size it. A full drum sends ester to VP-101 with no trip.	N3 N5 N7	N3.03 N5.01 N5.02 N7.06	9 Medium	Instrument / Safety
R23	Limit RY-104 entries (off-cut 10–25, product 8–20, assumed) and add R and HS-107 to the shift handover check.	N3 N5 N8	N3.05 N5.03 N5.05 N8.07	8 Medium	Control / Operations
R24	Fit XV-107 position switches with a discrepancy alarm against HS-107, and a WI-201 / WI-202 rate check.	N5 N6 N8	N5.05 N6.01 N8.07	8 Medium	Instrument
R25	Switch cuts on the D-105 lab result, not on TI-104 alone; flush the sample line; use a closed sample pot isolated from the vacuum.	N3 N6 N8	N3.06 N6.02 N6.08 N6.09 N8.06	8 Medium	Operations / Process
R26	Add a sample cooler at D-105 and a product cooler before TK-201 / TK-202; face shield and gloves for sampling and draining.	N5 N6 N8	N5.07 N6.07 N8.13	9 Medium	Safety / Process
R27	Add WI-201 / WI-202 high-weight alarms and size the receivers for the largest cut (Table 9 feed).	N6	N6.06	6 Medium	Process / Instrument
R28	Keep the receivers balanced to the vacuum system with a non-return valve; put XV-107 to CLOSED before any receiver is changed.	N6	N6.05	4 Low	Process / Operations
R29	Drain the residue only below 80 °C (assumed) through a closed drain; double-block the pot drain; do not charge a hot pot.	N1 N8	N1.09 N8.02 N8.13	9 Medium	Operations
R30	Before opening C-101: cool below 50 °C, purge with N ₂ and keep the packing wet; permit to work.	N3 N8	N3.07 N8.15	8 Medium	Safety / Operations
R31	Carry every High row and every row that relies on a SIS-100 trip into LOPA; record the SIL targets in the SRS.	N1 N2 N3 N4 N7 N8	N1.02 N1.03 N2.04 N3.08 N4.01 N4.02 N7.03 N7.06 N8.09 N8.11	12 High	Safety
R32	Alarm the operator at the cut switch and the end point (lab request reminders); staff night shifts of long campaigns.	N6 N8	N6.02 N6.03 N8.08	8 Medium	Operations
R33	Train the deviations of the dynamic check on the simulator (one per shift team) and revalidate this HAZOP every 5 years or at any MOC.	N1 N2 N3 N4 N5 N7 N8	N1.03 N1.05 N1.07 N2.01 N2.02 N2.04 N3.01 N3.03 N3.04 N3.08 N4.01 N4.03 N4.07 N5.01 N7.01 N7.03 N7.04 N7.05 N7.06 N8.11	12 High	Operations / Safety

5.4 Dynamic check on the simulator

Twelve deviations were run on copies of the model for 30 min each with all alarms and trips active and no operator action. Alarms were evaluated every second as on the DCS console (same limits, suppression and dead band). Times are mm:ss after the deviation starts.

Run	Deviation (node, start)	First alarms	Trip first-out	Range in 30 min
D1	Air in-leakage 12× (flange leak) (N7, product cut) Fault "air leak into the vacuum system" at full severity (60 s ramp)	0:20 C-101 AIR 0:29 PIC-102 H > 200 1:00 FIC-104 L < 20	no trip	PIC-102: 160–251 mbar PV-102: 0–50 % TI-106: 262.8–272.6 °C
D2	VP-101 vacuum pump trips (N7, product cut) Fault "VP-101 trips" at t = 0	0:57 FIC-104 L < 20	PSHH-102 at 9:31	PIC-102: 160–485 mbar TI-106: 262.8–273.3 °C JIC-101: 0.00–3.00 kW
D3	PV-102 N ₂ bleed fully open (N7, product cut) PIC-102 forced to MAN at 100 % (as PV-102 failing open)	0:10 PIC-102 H > 200 0:54 FIC-104 L < 20	no trip	PIC-102: 160–300 mbar TI-106: 262.8–278.0 °C TI-104: 175.2–266.6 °C
D4	Tempered-water pump P-108 trips (N4, product cut) Fault "P-108 trips" at t = 0	0:02 FI-108 L < 0.6 0:11 E-103 OVLD 1:14 FIC-104 L < 20	FSLL-108 at 0:14	FI-108: 0.00–1.20 m ³ /h PIC-102: 154–170 mbar Ester to trap: 0.00–1.19 kg
D5	Tempered water too cold (22 °C) (N4, product cut) Fault "tempered water too cold" at full severity (60 s ramp)	1:03 TI-109 L < 45	no trip	TI-109: 22.0–60.0 °C E-103 plugging: 0–59 % TI-105: 223–227 °C
D6	TT-107 reads 45 °C low (N2, total reflux) Fault "TT-107 reads low" at full severity (60 s ramp)	13:07 FIC-104 L < 20	TSHH-107 at 10:23	TIC-107 (TT-107): 249.2–290.0 °C Real oil (TE-107B): 290.0–330.2 °C JIC-101: 0.00–3.06 kW
D7	TV-101 wide open (JIC-101 MAN 100 %) (N2, total reflux) JIC-101 forced to MAN at 100 %	0:03 JIC-101 H > 5 0:10 PDI-103 H > 18 15:00 LIC-104 H > 85 19:09 E-103 OVLD	no trip	JIC-101: 3.0–12.0 kW Flooding: 52–136 % PDI-103: 11–142 mbar
D8	P-104 reflux pump trips (product cut) (N5, product cut) Fault "P-104 trips" at t = 0	3:14 E-103 OVLD 13:20 PIC-102 H > 200	no trip	LIC-104: 50–100 % PIC-102: 160–230 mbar Ester to trap: 0.00–13.13 kg
D9	Foaming in the pot (N1, total reflux) Fault "foaming in the pot" at full severity (60 s ramp)	9:42 PDI-103 H > 18	no trip	Flooding: 58–106 % PDI-103: 10.9–28.1 mbar PI-103: 171–188 mbar
D10	PV-102 closed (PIC-102 MAN 0 %) (N7, product cut) PIC-102 forced to MAN at 0 %	0:04 PDI-103 H > 18 0:06 PIC-102 L < 140	no trip	PIC-102: 20–160 mbar Flooding: 19–230 % TK-202 purity: 95.64–99.37 %
D11	P-101 hot-oil pump trips (N2, total reflux) Fault "P-101 trips" at t = 0	2:39 FIC-104 L < 20	no trip	JIC-101: 0.00–3.00 kW TI-106: 252.0–255.5 °C LIC-104: 41–51 %
D12	HV-110 opened with the pot hot (N8, product cut) HV-110 (N ₂ vacuum break) opened at t = 0 during the product cut	0:01 PIC-102 H > 200 0:59 FIC-104 L < 20	PSHH-102 at 0:06	PIC-102: 160–1013 mbar TI-106: 261.4–263.0 °C JIC-101: 0.00–3.00 kW

What the runs tell the team

D1 (rows N3.08 N7.03): Air in-leakage is the weakest-protected scenario. With a 12× leak the top pressure rises only to 251 mbar (PSHH-102 needs > 300), PV-102 closes to 0 % and the pot reaches 272.6 °C: no trip in 30 min. The only protection is the C-101 AIR alarm (0:20) and PIC-102 H (0:29) plus the operator. Take it to LOPA (R02, R31).

D2 (rows N7.01): A VP-101 trip is caught by PSHH-102 at 9:31, but the console suppresses the PIC-102 alarms while VP-101 is stopped, so the first alarm is FIC-104 L at 0:57. After the trip the top keeps rising to 485 mbar with the pot at up to 273.3 °C: the trip removes the heat, not the hot inventory (R05).

D3 (rows N7.05): With PV-102 failed open the top settles at 300.0 mbar, at the PSHH-102 setting (> 300 mbar): no trip in 30 min. Whether the plant trips depends on noise and calibration — the outcome is undefined (R07). First alarm: PIC-102 H at 0:10.

D4 (rows N4.01): Loss of tempered water: FI-108 L at 0:02, FSLL-108 trips at 0:14 (delay 10 s). 1.19 kg of ester still reaches the vacuum trap before boiling stops — size the trap and vent the exhaust outdoors (R06).

D5 (rows N4.03): Cold tempered water gives one alarm (TI-109 L at 1:03) and no trip in 30 min; E-103 plugging reaches 59 % in 30 min while the condensate TI-105 stays at 223 °C or more, so the TI-105 low alarm does not help. The operator must act on TI-109 alone (R19, R21).

D6 (rows N2.04): TT-107 reading low: TIC-107 shows 249–290 °C while the real oil reaches 330.2 °C. The TIC-107 high alarm never comes (same sensor); TSHH-107 on its own sensor TE-107B trips at 10:23. The independent sensor works, but only at the trip: a TT-107 / TE-107B deviation alarm (R08) would show the split within the first minute of the drift.

D7 (rows N2.02 N3.01): TV-101 wide open raises the duty to 12.0 kW, flooding to 136 % and ΔP to 142 mbar with no trip in 30 min; alarms JIC-101 H 0:03, PDI-103 H 0:10, LIC-104 H 15:00. Flooding has no automatic protection: R12 (BPCS clamp and ΔP override).

D8 (rows N5.01 N7.06): A P-104 trip in the product cut fills D-104 (100 %), raises the top to 230 mbar and sends 13.1 kg of ester to the vacuum trap in 30 min; no trip in 30 min. The LIC-104 alarms are suppressed while P-104 is stopped; the first alarm is E-103 OVLD at 3:14. R22 (LSHH-104 or condenser-overload trip) for LOPA.

D9 (rows N1.07 N3.04): Foam lifts flooding to 106 % and ΔP to 28.1 mbar (pot 188 mbar); PDI-103 H comes at 9:42, no trip in 30 min.

D10 (rows N7.04): PV-102 closed drops the top to 20 mbar: flooding 230 %, and the TK-202 composite purity falls from 99.37 to 95.64 % in 30 min — a quality loss of the whole receiver, no trip in 30 min. Alarms: PDI-103 H 0:04, PIC-102 L 0:06.

D11 (rows N2.01): A P-101 trip removes the duty (0.00 kW) and the pot cools to 252.0 °C: safe direction, no trip in 30 min, first alarm FIC-104 L at 2:39. HE-101 must stay interlocked to flow (R09).

D12 (rows N8.11): HV-110 opened with the pot hot: PSHH-102 trips heating at 0:06 and the column goes to 1013 mbar, but the pot never falls below 261 °C in the 30 min. The trip cannot make an early vacuum break safe; only the HV-110 permissive (R03) can.

5.5 Follow-up

1. Each recommendation gets an owner, a target date and a written response on its close-out sheet (Appendix 3); status starts OPEN.
2. LOPA for the 10 rows of 5.2, including the proposed functions for air ingress (R02), D-104 high-high (R22) and the HV-110 permissive (R03).
3. Management of change reopens the affected node for any change of set points, trip settings, alarm limits, the as-built options or the feed.
4. Revalidation every 5 years or after an incident; simulator training on the Part 9 deviations (R33).

Appendix 1 · HAZOP worksheets

N1 · Pot D-101 and reboiler D-102 (process side)

Project	Node	Design intent	Design conditions	Equipment	Instruments	Drawing
Vacuum Batch Distillation Ver 1 (pilot column C-101)	N1 Pot D-101 and reboiler D-102 (process side)	Hold the 110 L charge (95.7 kg) in D-101 and boil it at pot pressure 171 mbar with 3.0 kW from D-102, sending the vapour into C-101; the residue (C18 + C20 rich) stays in the pot at the end.	Pot 255.5 °C, 171 mbar, level 89 % at the start of the campaign	D-101 pot (150 L), D-102 reboiler (process side), drain valve	TIC-101 (TT-106, pot temperature), TE-101B / TSHH-101, LI-101 / LSLL-101 (LT-101), PI-103, TI-101	VBD-V1-PID-001 rev A (screen P&ID)

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N1.01	Level	MORE	Pot overfilled at charging (above the 110 L charge)	- Wrong charge volume entered or measured - charge from a second drum added - LT-101 reads low	Liquid close to the column bottom; residue entrained into the packing at first boil; heavies (C18) in the off-cut and product; foam-over at heat-up	- LI-101 indication only (no high alarm) - charging procedure 110 L	2	3	6 Medium	R15	Instrument, Operations
N1.02	Level	LESS	Pot boiled down at the end of the campaign	- Product cut not stopped at the end point - richer feed (Table 9 feed, C16:0 58.6 %) distils more - LT-101 reads high	D-102 surface exposed; pot overheats; ester degradation and polymer on D-102 (fouling for later batches); TSHH-101 demand	- LI-101 L alarm 30 % - LSLL-101 trips heating at < 20 % (5 s, shares LT-101 with LI-101) - TSHH-101 (TE-101B) at > 290 °C - end-of-campaign rule (cumulative TK-202 purity) - SIF-01 TSHH-101, SIF-04 LSLL-101	3	3	9 Medium	R13 R14 R31	Instrument, Safety, Operations, Process
N1.03	Temperature	MORE	Pot temperature high (above the TIC-101 alarm)	- Heavies concentrate late in the cut - top pressure high (vacuum loss, N7) - hot oil above 290 °C (N2) - duty set too high	Thermal degradation of the esters above ≈ 280 °C (assumed limit), dark residue, polymer on D-102, TSHH-101 trip and lost batch time	- TIC-101 H alarm 280 °C - TSHH-101 trips heating at > 290 °C on its own sensor TE-101B (5 s) - hot oil held at 290 °C by TIC-107 (pot cannot exceed the oil temperature) - SIF-01 TSHH-101	3	3	9 Medium	R14 R18 R31 R33	Operations, Process, Safety
N1.04	Temperature	LESS	Pot does not reach boiling / heat-up very slow	- HE-101 heater elements burnt out - D-102 fouled - P-101 tripped - TV-101 failed closed (air loss)	No boil-up, batch delayed; operator raises the duty set point and floods the column when the cause clears	- JIC-101 duty and TV-101 output on the DCS - TI-101 and TI-108 show the film ΔT	1	3	3 Low		
N1.05	Pressure	MORE	Pot pressure high (PI-103 rises)	- Column flooding (ΔP up to the packing limit) - top pressure high (N7) - foaming	Pot boiling point rises with pressure; pot temperature and degradation rise; less boil-up for the same duty	- PDI-103 H alarm 18 mbar - PIC-102 H alarm 200 mbar - PSHH-102 trips heating at > 300 mbar (top) - SIF-02 PSHH-102	2	3	6 Medium	R12 R33	Control, Operations, Safety
N1.06	Composition	OTHER THAN	Wrong or wet feed charged (water, methanol traces, other FAME)	- Biodiesel not dried or methanol not stripped upstream - charge taken from the wrong drum - Table 9 feed charged when the design feed was planned	Water or methanol flash at heat-up: pressure surge, foam, flammable vapour to the VP-101 exhaust; with a different feed the cut points move and fixed cut times give off-spec product	- PIC-102 H alarm 200 mbar - slow heat-up procedure - lab sample of the charge (not formalised)	3	2	6 Medium	R16 R06	Operations, Laboratory, Process

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N1.07	Composition	AS WELL AS	Foam in the pot (surface-active degradation products)	- Residue from the previous batch left in the pot - ester degraded by over-heating - duty raised quickly	Flooding index rises by up to 60 %, heavies carried into the packing and product, ΔP high	- PDI-103 H alarm 18 mbar - JIC-101 H alarm 5 kW - flooding index on the DCS	2	3	6 Medium	R17 R12 R33	Operations, Control, Safety
N1.08	Heat duty	LESS	Reboiler D-102 fouled (polymer on the heating surface)	- Pot over-heated above ≈ 285 °C in earlier batches - residue not fully drained - no cleaning interval	UA falls (up to –95 %); TV-101 runs wide open; duty cannot be held; longer campaign; hotter film makes more polymer	- JIC-101 duty with TV-101 output - TI-101 (reboiler outlet) minus TI-106 shows the film ΔT	2	3	6 Medium	R18	Process
N1.09	Containment	OTHER THAN	Hot ester released at the pot drain or flange	- Drain valve opened or passing - gasket failure after thermal cycling - vacuum broken so the pot is no longer below atmospheric	Ester at ≈ 255–270 °C (above its flash point, assumed) spills: burns, pool fire	- Operation under vacuum (leaks go inward while under vacuum) - drain valve closed by procedure	4	2	8 Medium	R29 R10	Operations, Safety, Mechanical
N1.10	Flow	REVERSE	Hot oil leaks into the process side (D-102 tube leak)	Tube or coil leak in D-102 (oil side at a higher pressure than the pot under vacuum)	Hot oil contaminates residue and lights; cracked oil vapour to the vacuum system; hot-oil inventory falls (N2)	- None direct - residue colour / lab sample	2	2	4 Low	R09	Process, Mechanical

N2 · Hot-oil loop (HE-101, P-101, XV-101, TV-101, E-104)

Project	Node	Design intent	Design conditions	Equipment	Instruments	Drawing
Vacuum Batch Distillation Ver 1 (pilot column C-101)	N2 Hot-oil loop (HE-101, P-101, XV-101, TV-101, E-104)	Circulate hot oil at 290 °C (TIC-107 moves HE-101, 12 kW) with P-101 through XV-101 and TV-101 to D-102 so that the pot receives the duty set on JIC-101 (3.0 kW at design, TV-101 ≈ 40 %); E-104 / HV-104 cools the oil only at cool-down.	Supply 290 °C, return 286 °C, duty 3.0 kW	HE-101 electric heater (12 kW), P-101 hot-oil pump, XV-101 SIS valve (fail closed), TV-101 duty valve (fail closed), D-102 (oil side), E-104 cooler with HV-104	TIC-107 (TT-107), TE-107B / TSHH-107, JIC-101, TI-108 (return), TI-101; TIC-101 on/off of P-101 in the as-built option	VBD-V1-PID-001 rev A (screen P&ID)

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N2.01	Flow	NO	No hot-oil flow to D-102	- P-101 trips (motor, coupling) - XV-101 closed (trip or spurious) - TV-101 fails closed (air loss) - as-built TIC-101 stops P-101	Boil-up stops (safe direction); batch time lost; the heater must not run without flow or the oil in HE-101 overheats locally	- HE-101 is enabled only with P-101 running (model, to be confirmed on the plant) - JIC-101 duty and TI-108 on the DCS	2	3	6 Medium	R09 R33	Process, Mechanical, Operations, Safety
N2.02	Flow	MORE	TV-101 wide open: duty far above the 3 kW set point	- JIC-101 left in MAN at a high output - heat-up set point (up to 8 kW) not cut before boiling - TV-101 positioner fault	Duty rises to ≈ 12 kW; column floods, ΔP very high, D-104 fills, vapour reaches VP-101; no trip acts	- JIC-101 H alarm 5 kW - PDI-103 H alarm 18 mbar - LIC-104 H alarm 85 %	2	4	8 Medium	R12 R33	Control, Operations, Safety
N2.03	Flow	PART OF	Duty arrives in pulses (as-built on/off heating)	As-built TIC-101 on/off of P-101 (±1 °C band) used instead of JIC-101	Duty swings between zero and the full loop capacity: flooding bursts, ΔP spikes, poor separation	- PDI-103 H alarm 18 mbar - JIC-101 H alarm 5 kW	2	4	8 Medium	R12	Control, Operations

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N2.04	Temperature	MORE	Hot oil above 290 °C	• TT-107 reads low (loose in its thermowell) so TIC-107 drives HE-101 up • TIC-107 in MAN at a high output • heater contactor welded	Oil degradation (coke, acids); oil above its flash point (assumed) so any leak can ignite; hotter D-102 film, pot degradation	• TIC-107 H alarm 315 °C (uses TT-107, so blind when TT-107 reads low) • TSHH-107 trips heating at > 330 °C on its own sensor TE-107B (5 s) • TI-108 return temperature SIF-05 TSHH-107	4	2	8 Medium	R08 R31 R33	Instrument, Safety, Operations
N2.05	Temperature	LESS	Hot oil too cold to hold the duty	• Half of the HE-101 elements burnt out (up to –60 %) • E-104 / HV-104 left open during the run • TIC-107 set point low	TV-101 runs wide open, duty below set point, longer campaign; the operator may raise R or cut time to compensate	• TIC-107 indication • JIC-101 output at 100 % visible on the DCS	1	3	3 Low	R11	Control
N2.06	Pressure	MORE	Hot-oil loop over-pressured	• TV-101 and XV-101 both closed with P-101 running (dead-head) • expansion line blocked • water in new oil flashing at first heat-up	Gasket or seal failure: hot oil at 290 °C sprayed (fire, burns)	• Expansion tank and relief on the loop (assumed — not shown on the screen P&ID) • TV-101 closes only on trip or by the operator	4	2	8 Medium	R09	Process, Mechanical
N2.07	Containment	OTHER THAN	Hot-oil leak at a flange or the P-101 seal	• Gasket creep from thermal cycling every batch • P-101 seal wear • flange bolts not re-torqued after maintenance	Hot oil at 290 °C sprays or pools: oil fire, severe burns	• HS-100 emergency stop (stops heating, XV-101 closes) • insulation • inspection by operators	4	2	8 Medium	R10 R09	Safety, Mechanical, Process
N2.08	Flow	REVERSE	Cooling water enters the hot oil through an E-104 tube leak	• E-104 tube leak while the cooling side pressure is higher than the oil	Water flashes in 290 °C oil: pressure surge in the loop, oil foams, leaks	• HV-104 kept closed during the run (cool-down only)	4	1	4 Low	R09	Process, Mechanical
N2.09	Composition	AS WELL AS	Air or water in the hot-oil loop	• Loop not purged after maintenance • open expansion tank • wet new oil	Oil oxidation and sludge, reduced heat transfer; water flashing at heat-up	• Start-up procedure (slow heat-up of the oil)	2	2	4 Low	R09	Process, Mechanical

N3 · Column C-101 (packing, profile, ΔP)

Project	Node	Design intent	Design conditions	Equipment	Instruments	Drawing
Vacuum Batch Distillation Ver 1 (pilot column C-101)	N3 Column C-101 (packing, profile, ΔP)	Separate the vapour from the pot over 20 equilibrium stages (structured packing, Murphree E 0.6) at 160 mbar top, ΔP 11 mbar and flooding index 61 % so that the light esters and then C16:0 leave at the top.	TI-104 192.4 °C, TI-103 193.2 °C, TI-102 237 °C at total reflux; ΔP 10.9 mbar	C-101 structured packing, liquid distributor, packing support	PDI-103, PI-103, TI-104 (stage 6), TI-103 (stage 13), TI-102 (stage 20), flooding index (calculated)	VBD-V1-PID-001 rev A (screen P&ID)

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N3.01	Flow	MORE	Boil-up too high: column floods	• Duty set point too high or TV-101 wide open (N2) • foam (N1) • heat-up duty not cut before boiling • top pressure too low (N7)	Flooding index above 100 %, ΔP rises steeply, stage efficiency falls, heavies reach the top: off-spec off-cut or product; liquid carried to E-103	• PDI-103 H alarm 18 mbar • JIC-101 H alarm 5 kW • flooding index display • LIC-104 H alarm 85 %	2	4	8 Medium	R12 R33	Control, Operations, Safety

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N3.02	Flow	LESS	Boil-up too low: weeping, poor wetting	• HE-101 elements failed • D-102 fouled • hot oil too cold (N2)	Below ≈ 25 % flooding the packing weeps and efficiency falls; cut takes longer; product purity slightly lower	• JIC-101 indication • temperature profile	1	3	3 Low		
N3.03	Flow	NO	No reflux to the packing	• P-104 trips • LSLL-104 stops P-104 • drum emptied by a high-reading LT-104	Packing dries from the top; top temperature rises; heavies reach the drum; drum and E-103 fill (N5)	• FIC-104 L alarm 20 kg/h • LIC-104 L/H alarms 20 / 85 %	2	3	6 Medium	R22 R33	Instrument, Safety, Operations
N3.04	Pressure	MORE	Column ΔP high	• Flooding • foam • packing fouled with polymer or crushed	Pot pressure = top + ΔP rises; pot temperature and degradation rise; efficiency falls	• PDI-103 H alarm 18 mbar	2	3	6 Medium	R12 R17 R33	Control, Operations, Safety
N3.05	Temperature	MORE	Heavies rise in the column (top temperature high)	• Reflux ratio too low (wrong entry in RY-104) • duty high • C18 front reaches the top late in the product cut • packing maldistribution	C18 in TK-202: composite purity below 99 %, whole receiver off-spec	• TI-104 and TI-103 trends • lab sample at D-105 • end-of-campaign rule	2	4	8 Medium	R14 R23	Operations, Process, Control
N3.06	Temperature	LESS	TI-104 reads low (measurement)	• TT-104 loose in its thermowell (reads up to 25 °C low)	The top looks "clean" too early: switch to product while lights are still present; lights in TK-202	• TI-103 for comparison • lab sample of D-105 before switching	2	3	6 Medium	R25	Operations, Process
N3.07	Composition	PART OF	Fewer effective stages than designed	• Packing maldistribution or damage (efficiency down by up to 60 %) • liquid distributor partly plugged with frozen ester or polymer	Purity of the product cut below 99 %; smaller on-spec cut; lower recovery	• Temperature profile TI-104 / TI-103 / TI-102 • lab results	2	3	6 Medium	R30	Safety, Operations
N3.08	Composition	AS WELL AS	Air enters the hot column (in-leakage)	• Flange, thermowell or sight-glass seal leak (air in-leakage up to 12x normal) • sample valve opened under vacuum	Hot ester (≈ 255–270 °C) and air at sub-atmospheric pressure: autoignition or fire in the packing or the vapour line	• PIC-102 H alarm 200 mbar (PV-102 first closes and hides a small leak) • C-101 AIR alarm on the DCS • PSHH-102 at > 300 mbar only for a large leak - SIF-02 PSHH-102	4	3	12 High	R01 R02 R04 R31 R33	Operations, Process, Instrument, Safety, Control

N4 · Condenser E-103 and tempered water (P-108, HV-105)

Project	Node	Design intent	Design conditions	Equipment	Instruments	Drawing
Vacuum Batch Distillation Ver 1 (pilot column C-101)	N4 Condenser E-103 and tempered water (P-108, HV-105)	Condense all overhead vapour at 160 mbar in E-103 with tempered water at 60 °C and 1.2 m³/h; keep the water above 45 °C so that C16/C18 esters (m.p. 30–38 °C) do not freeze on the tubes.	Water 60 °C, 1.20 m³/h; condensate TI-105 176 °C at total reflux	E-103 condenser, P-108 tempered-water pump, HV-105 (HIC-105) with strainer, tempered-water heater/TCV (outside the screen P&ID)	FI-108 / FSLL-108 (FT-108), TI-109, TI-105, HIC-105	VBD-V1-PID-001 rev A (screen P&ID)

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N4.01	Flow	NO	No tempered water to E-103	- P-108 trips (motor, power) - HV-105 closed by mistake - loop isolation valve shut	Vapour no longer condensed: ester vapour to VP-101 and the vent, top pressure rises; no reflux	- FI-108 L alarm 0.6 m³/h - FSLL-108 trips heating at < 0.36 m³/h (10 s, shares FT-108 with FI-108) - PSHH-102 at > 300 mbar - SIF-02 PSHH-102, SIF-03 FSLL-108	3	3	9 Medium	R05 R06 R31 R33	Mechanical, Instrument, Process, Safety, Operations
N4.02	Flow	LESS	Tempered-water flow low	- HV-105 strainer blocking (flow down by up to 80 %) - HIC-105 set too low - P-108 worn	Condenser capacity falls; part of the vapour passes to VP-101; FSLL-108 trips heating when the flow falls below its setting	- FI-108 L alarm 0.6 m³/h - FSLL-108 at < 0.36 m³/h (10 s) - SIF-03 FSLL-108	3	3	9 Medium	R20 R31	Mechanical, Safety
N4.03	Temperature	LESS	Tempered water too cold (below 45 °C)	- Tempered-water TCV fails (supply falls to ≈ 22 °C) - loop heater off - cold water used at start-up	C16/C18 esters freeze on the tubes: E-103 plugs, condenser capacity and vacuum lost, column upset, ester to VP-101	- TI-109 L alarm 45 °C - TI-105 L alarm 45 °C - no trip	3	3	9 Medium	R19 R21 R33	Instrument, Process, Operations, Safety
N4.04	Temperature	MORE	Tempered water too hot (above 80 °C)	Loop heater or TCV fails towards heating	Less temperature difference: condenser capacity falls a little (dew point ≈ 190 °C); small effect	TI-109 H alarm 80 °C	1	2	2 Low		
N4.05	Flow	REVERSE	Water enters the process side (E-103 tube leak)	- Tube leak - process side is under vacuum so water leaks inward	Water flashes in E-103/D-104: pressure rises, P-104 cavitates, wet product	- PIC-102 H alarm 200 mbar - lab sample of the distillate	2	2	4 Low	R16	Operations, Laboratory
N4.06	Phase	AS WELL AS	Frozen ester left in E-103 from the last batch	- Tempered water left running cold after shutdown - cold start with water below 45 °C	Next start-up with a partly plugged condenser: high pressure, vapour to VP-101	- TI-109 L alarm 45 °C - start-up procedure	2	3	6 Medium	R21	Operations
N4.07	Heat duty	MORE	Vapour load above the condenser capacity	- Duty much higher than design (N2/N3) - condenser capacity margin 2.5× the design load (assumed)	Uncondensed vapour to VP-101, pressure rises	- E-103 OVLD alarm (condenser overloaded) on the DCS - JIC-101 H alarm 5 kW	2	3	6 Medium	R12 R06 R33	Control, Operations, Process, Safety

N5 · Reflux drum D-104, P-104, FV-104 / FV-105

Project	Node		Design intent		Design conditions	Equipment	Instruments				Drawing
Vacuum Batch Distillation Ver 1 (pilot column C-101)	N5 Reflux drum D-104, P-104, FV-104 / FV-105		Hold D-104 (4 L) at 50 % with LIC-104 and return reflux through FV-104 (fail open) at the ratio set on RY-104 (R 15 off-cut, R 10 product); the distillate leaves through FV-105 (fail closed).		Drum 51 %, reflux 32 kg/h at total reflux; product cut: reflux 34.9 kg/h, distillate 3.49 kg/h	D-104 reflux drum, P-104 reflux / distillate pump, FV-104 (FO), FV-105 (FC)	LIC-104 (LT-104), LSLL-104 (pump protection), RY-104, FIC-104 (FT-104), FIC-105 (FT-105)				VBD-V1-PID-001 rev A (screen P&ID)
ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N5.01	Level	MORE	D-104 fills up	• P-104 trips • LIC-104 left in MAN at a low output • LT-104 reads low • FV-104 stuck closed	Drum full: condensate backs up into E-103, condenser area lost, ester vapour to VP-101 and the trap; top pressure rises; no reflux, packing dries; no trip acts	• LIC-104 H alarm 85 % • FIC-104 L alarm 20 kg/h • PIC-102 H alarm 200 mbar • E-103 OVLD alarm	3	3	9 Medium	R22 R06 R33	Instrument, Safety, Process, Operations
N5.02	Level	LESS	D-104 runs dry	• LT-104 drifts high (up to +40 % of span) so LIC-104 draws the drum down • FT-105 reads low (more drawn than shown) • duty lost	P-104 runs dry (seal and impeller damage), reflux lost; LSLL-104 stops P-104 and the case then becomes "drum fills"	• LIC-104 L alarm 20 % • LSLL-104 stops P-104 at < 5 % (5 s, equipment interlock, not a SIF) • P-104 DRY alarm	2	3	6 Medium	R22	Instrument, Safety
N5.03	Reflux ratio	LESS	Real reflux ratio below the RY-104 value	• FT-104 reads high (up to +40 %) • FV-104 sticks • R typed as 3 instead of 10 at a shift change	Heavies (product cut) or C16 (off-cut) leave at the top: product off-spec or C16:0 lost to the off-cut	• Lab sample at D-105 • TI-104 / TI-103 trends • FIC-104 L alarm 20 kg/h	2	4	8 Medium	R23	Control, Operations
N5.04	Reflux ratio	MORE	Reflux far above the set ratio (no draw)	• FV-104 fails open (air loss) • FIC-104 in MAN at a high output	No distillate; drum level falls; batch time lost (safe direction)	• LIC-104 L alarm 20 % • FIC-105 indication	1	3	3 Low		
N5.05	Flow	MORE	Distillate rate too high	• FT-105 reads low (up to –40 %) • FIC-105 in MAN at a high output • R too low	Purity of the cut falls; drum drawn down	• LIC-104 L alarm 20 % • WI-201 / WI-202 rate • lab sample	2	3	6 Medium	R24 R23	Instrument, Control, Operations
N5.06	Flow	NO	No distillate	• FV-105 fails closed (air loss) • XV-107 CLOSED • P-104 stopped	No product; campaign longer (safe direction)	• FIC-105 and WI-201 / WI-202 indication	1	3	3 Low		
N5.07	Containment	OTHER THAN	Hot ester leak at the P-104 seal	• Seal wear • dry running (above) • under vacuum the seal also draws air in	Ester at ≈ 180-230 °C sprayed: burns, fire; air drawn into the hot system	• Operator rounds • HS-100	3	2	6 Medium	R26 R01	Safety, Process, Operations
N5.08	Composition	AS WELL AS	Water or methanol in the drum at start-up	• Wet or unstripped charge (N1) • E-103 tube leak (N4)	P-104 vapour-locks or cavitates; pressure swings; flammable vapour to VP-101	• PIC-102 H alarm 200 mbar	2	2	4 Low	R16	Operations, Laboratory

N6 · Distillate routing XV-107 / HS-107, D-105, TK-201, TK-202

Project	Node	Design intent	Design conditions	Equipment	Instruments	Drawing
Vacuum Batch Distillation Ver 1 (pilot column C-101)	N6 Distillate routing XV-107 / HS-107, D-105, TK-201, TK-202	Send no distillate at total reflux, the off-cut (C8-C14) to TK-201 at R 15 and the product (C16:0 $\geq$ 99 wt %) to TK-202 at R 10, switching on the D-105 sample; stop the product cut before the C18 front.	Reference campaign: switch at 307 min, end at 907 min, TK-201 11.1 kg, TK-202 34.7 kg at 99.0 % C16:0	XV-107 three-way diverter, D-105 sample point, TK-201 off-cut receiver, TK-202 product receiver	HS-107, WI-201, WI-202, lab GC of D-105	VBD-V1-PID-001 rev A (screen P&ID)

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N6.01	Routing	OTHER THAN	Distillate goes to the wrong receiver	- HS-107 set to the wrong position - XV-107 fails to move (HS-107 changes the indication only) - receivers swapped after maintenance	Lights in TK-202 (the whole product receiver off-spec) or product lost to TK-201	- WI-201 / WI-202 weights (which tank is rising) - lab sample	2	4	8 Medium	R24	Instrument
N6.02	Routing	EARLY	Switch to product too early	- TI-104 reads low - switch made on time instead of the sample - lab result read from an earlier sample	Light esters in TK-202: composite purity below 99 %	- Switch rule (D-104 light esters < 4 %) - lab GC of D-105 ($\approx$ 10 min delay)	2	4	8 Medium	R25 R32	Operations, Process
N6.03	Routing	LATE	Product cut stopped too late	- End point missed (night shift, lab delay, operator away) - TI-103 rise not noticed	C18 front enters TK-202: cumulative purity falls below 99 %, the whole receiver off-spec	- TI-103 / TI-104 trends - lab sample - WI-202	2	4	8 Medium	R14 R32	Operations, Process
N6.04	Routing	BEFORE	Distillate drawn before total reflux has lined out	- Off-cut started before the profile is established - FIC-105 put in AUTO too early	Light esters spread through the column: longer off-cut, C16:0 lost to TK-201	Procedure: line out at total reflux first	1	3	3 Low		
N6.05	Flow	REVERSE	Back-flow from a receiver into the column or drum	- Receiver vented to atmosphere or N₂ while the column is under vacuum - XV-107 passing when P-104 stops	Off-cut or product sucked back; N ₂ or air drawn into the hot drum and column	- Receivers on the vacuum system (assumed) - XV-107 CLOSED at total reflux	2	2	4 Low	R28	Process, Operations
N6.06	Level	MORE	Receiver overfilled	- Receiver smaller than the cut (Table 9 feed gives a larger product cut) - weight not watched	Hot ester overflows: burns, fire risk	WI-201 / WI-202 weights (no alarm)	3	2	6 Medium	R27	Process, Instrument
N6.07	Temperature	MORE	Hot distillate at the sample point and receivers	No product cooler: distillate at $\approx$ TI-105 (227 °C in the product cut) reaches D-105 and the receivers	Burns when sampling or changing receivers; ester vapour at the sample point	PPE and sampling procedure	3	3	9 Medium	R26	Safety, Process
N6.08	Composition	PART OF	Sample at D-105 not representative	Sample line not flushed (dead leg holds an earlier cut)	Wrong switch or stop decision	Sampling procedure	2	3	6 Medium	R25	Operations, Process
N6.09	Containment	AS WELL AS	Air drawn in through the D-105 sample valve	Sample valve opened while D-105 is under vacuum	Air into hot condensate and the column (as N3 / N7)	Sampling procedure (isolate before opening)	4	2	8 Medium	R25 R02	Operations, Process, Instrument, Safety

N7 · Vacuum system (VP-101, trap, PV-102 N₂ bleed, HV-110)

Project	Node	Design intent	Design conditions	Equipment	Instruments	Drawing
Vacuum Batch Distillation Ver 1 (pilot column C-101)	N7 Vacuum system (VP-101, trap, PV-102 N ₂ bleed, HV-110)	Hold the column top at 160 mbar with VP-101 and the PV-102 N ₂ bleed (fail open, ≈ 50 % at design) against an air in-leakage of 20 (model units); collect uncondensed ester in the trap; HV-110 breaks vacuum with N ₂ only when the pot is below 150 °C.	Top 160 mbar, pot 171 mbar; PSHH-102 active only while heating is armed	VP-101 vacuum pump with trap and exhaust, PV-102 N ₂ bleed valve (FO), HV-110 N ₂ vacuum break, N ₂ supply	PIC-102 (PT-102), PSHH-102 (shares PT-102), PI-103	VBD-V1-PID-001 rev A (screen P&ID)

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N7.01	Pressure	MORE	Vacuum lost: VP-101 trips	- Motor or power failure - overload from ester condensed in the pump oil	Top pressure rises; boiling temperatures rise; PSHH-102 trips heating; after the trip the pressure keeps rising while the column is still hot	- PIC-102 H alarm 200 mbar - PSHH-102 trips heating at > 300 mbar (5 s, shares PT-102 with PIC-102) - SIF-02 PSHH-102	2	3	6 Medium	R05 R33	Mechanical, Instrument, Operations, Safety
N7.02	Pressure	MORE	Slow vacuum loss: VP-101 capacity falls	- Worn vanes - pump oil diluted by condensed ester	PV-102 closes to compensate, then pressure creeps up; hotter pot; less spare capacity to hide a leak	- PIC-102 H alarm 200 mbar - PV-102 output on the DCS	2	3	6 Medium	R04 R06	Control, Process
N7.03	Composition	AS WELL AS	Air in-leakage into the hot system	- Flange or seal leak (up to 12× normal in-leakage) - thermowell or sight-glass fitting - P-104 seal under vacuum	Air meets hot ester (≈ 255–270 °C) in the column and vapour line: autoignition or fire; pressure only partly rises and the trip is not reached	- PIC-102 H alarm 200 mbar - C-101 AIR alarm on the DCS - PSHH-102 at > 300 mbar (not reached by a 12× leak) - SIF-02 PSHH-102	4	3	12 High	R01 R02 R04 R31 R33	Operations, Process, Instrument, Safety, Control
N7.04	Pressure	LESS	Top pressure too low	- PIC-102 in MAN with PV-102 closed - PT-102 reads high - N₂ bleed isolated	Vapour velocity rises at low pressure; column floods, ΔP very high, product off-spec	- PIC-102 L alarm 140 mbar - PDI-103 H alarm 18 mbar	2	3	6 Medium	R12 R07 R33	Control, Operations, Process, Instrument, Safety
N7.05	Flow	MORE	N ₂ bleed too high: PV-102 fails open	- Instrument-air failure (PV-102 fails open) - PIC-102 in MAN at a high output	Top pressure settles near the PSHH-102 setting: hot pot, slower separation; trip may or may not come (spurious trips)	- PIC-102 H alarm 200 mbar - PSHH-102 at > 300 mbar - SIF-02 PSHH-102	2	3	6 Medium	R07 R33	Process, Instrument, Operations, Safety
N7.06	Flow	MORE	Uncondensed ester to VP-101	- Condenser cooling lost (N4) - drum full (N5) - flooding (N3)	Ester in the pump oil (VP-101 capacity loss); ester mist at the exhaust (flammable vapour indoors); trap fills	- Trap ahead of VP-101 - E-103 OVLD alarm - FSLL-108 trip for loss of cooling - SIF-03 FSLL-108	3	2	6 Medium	R06 R22 R31 R33	Process, Instrument, Safety, Operations
N7.07	Composition	OTHER THAN	Methanol traces and light esters at the VP-101 exhaust	- Unstripped methanol in the charge - lights flashed at heat-up	Flammable vapour indoors near the pump	- Trap - general ventilation of the pilot hall	3	2	6 Medium	R06 R16	Process, Operations, Laboratory
N7.08	Flow	REVERSE	Back-flow from VP-101 into the column when the pump stops	VP-101 stops without a non-return or isolation valve on its suction	Pump oil mist and air drawn back into the hot column	None on the screen P&ID	3	2	6 Medium	R05	Mechanical, Instrument
N7.09	Composition	OTHER THAN	Air instead of N ₂ at PV-102 or HV-110	- N₂ cylinder empty and swapped for air - wrong hose connected	The bleed or the vacuum break feeds air into the hot column (as the air in-leakage row)	N₂ supply procedure	4	2	8 Medium	R03 R02	Instrument, Process, Safety

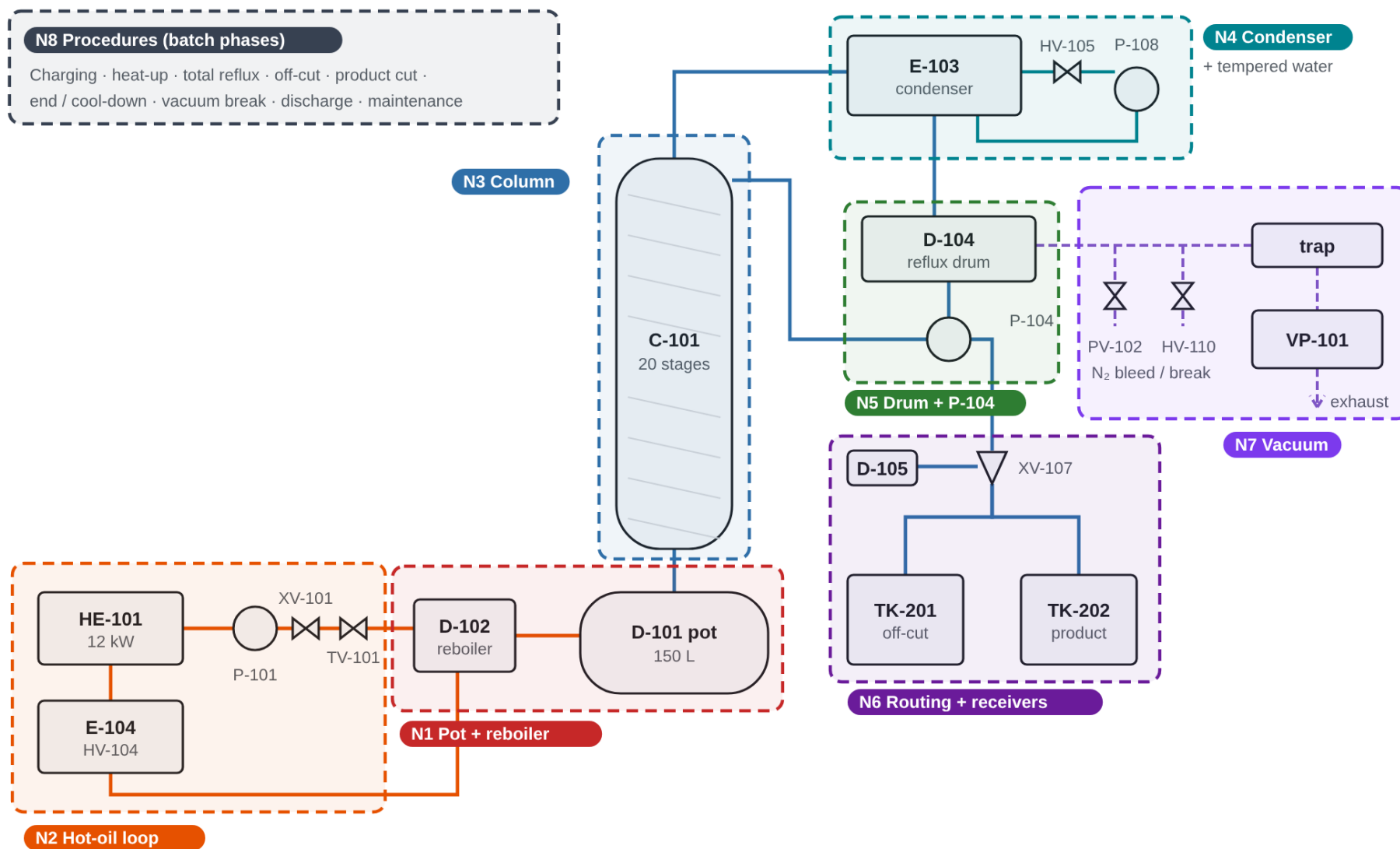
N8 · Procedures (batch phases)

Project	Node		Design intent		Design conditions	Equipment	Instruments				Drawing
Vacuum Batch Distillation Ver 1 (pilot column C-101)	N8 Procedures (batch phases)		Run the batch safely through each phase: charging, heat-up, total reflux, off-cut, product cut, end and cool-down, vacuum break, discharge of the residue, and maintenance.		Reference campaign A: off-cut 307 min, end 907 min; cool-down to < 150 °C before the vacuum break	Whole unit	SIS-100 arming permissives, HS-100, the five procedures of the page (start-up, campaign A and B, shutdown, loss of cooling)				VBD-V1-PID-001 rev A (screen P&ID)
ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N8.01	Charging	OTHER THAN	Different feed charged than planned	- Table 9 feed (C16:0 58.6 %) or another lot charged - no lab analysis	Cut points move: a cut plan fixed in time gives off-spec product or lost C16:0	Lab sample of D-105 during the cuts	2	3	6 Medium	R16	Operations, Laboratory
N8.02	Charging	BEFORE	Charging into a pot still hot from the last batch	- Turn-around shortened - residue not cooled	Fresh ester and air on hot surfaces: vapour release, fire	Cool-down procedure	4	2	8 Medium	R29	Operations
N8.03	Heat-up	BEFORE	Heating armed before vacuum and cooling are established	Operator out of sequence	Boiling without condenser or vacuum: vapour release	SIS-100 arming permissives: VP-101 running and PIC-102 < 250 mbar, P-108 running and FI-108 ≥ 0.6 m³/h, LI-101 > 30 %, P-101 running, TIC-107 < 320 °C, HV-110 closed	3	1	3 Low		
N8.04	Heat-up	MORE	Heat-up duty not cut before the first boil	Heat-up set point (up to 8 kW) left on JIC-101	Column floods at the first boil; heavies to the top; vapour to VP-101	- JIC-101 H alarm 5 kW - PDI-103 H alarm 18 mbar	2	4	8 Medium	R12	Control, Operations
N8.05	Total reflux	PART OF	P-104 started before D-104 has enough liquid	Pump started too early	Dry running of P-104; LSLL-104 stops it	- LSLL-104 at < 5 % (5 s) - procedure: start P-104 when D-104 > 30 %	2	3	6 Medium		
N8.06	Off-cut	AFTER	Decision taken on a lab result that arrives after the change it should guide	- GC result ≈ 10 min after sampling - sample taken late	Switch to product late (C16:0 lost) or early (lights in product)	- Switch rule - TI-104 trend	2	3	6 Medium	R25	Operations, Process
N8.07	Product cut	AS WELL AS	Shift handover during the product cut	- Wrong R entered (3 instead of 10) - HS-107 position not handed over	Product off-spec or sent to the wrong receiver	- Lab sample - WI-202	2	4	8 Medium	R23 R24	Control, Operations, Instrument
N8.08	Product cut	NO	Nobody watching the end of a 15 h campaign	- Long campaign over a night shift - operator away	End point missed: C18 into TK-202, pot boiled down	- LI-101 L alarm 30 % - LSLL-101 at < 20 % - SIF-04 LSLL-101	2	4	8 Medium	R32 R14	Operations, Process
N8.09	End / cool-down	LATE	Heating not stopped at the end of the campaign	Distillate closed but TIC-107 and JIC-101 left in AUTO	Pot held hot with a small heavy residue: degradation, polymer on D-102, TSHH-101 demand	- TIC-101 H alarm 280 °C - TSHH-101 at > 290 °C (TE-101B, 5 s) - LSLL-101 at < 20 % - SIF-01 TSHH-101, SIF-04 LSLL-101	3	3	9 Medium	R14 R31	Operations, Process, Safety
N8.10	End / cool-down	LESS	Cool-down too slow	- HV-104 not opened - P-101 stopped too early (no circulation through D-102)	Long wait under vacuum; pressure to break vacuum early	Shutdown procedure (cool through D-102 and E-104)	2	3	6 Medium	R03	Instrument, Process

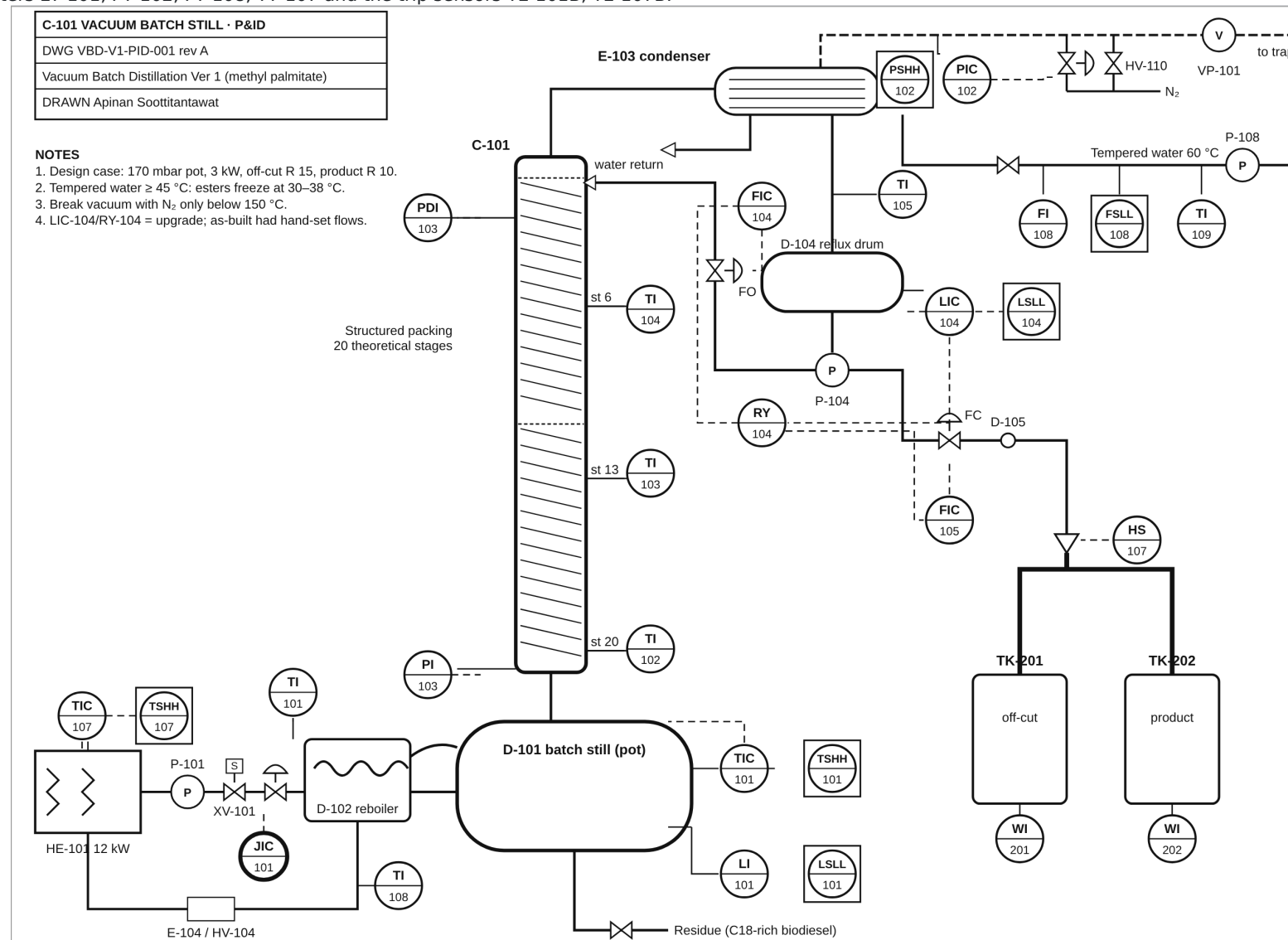
ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N8.11	Vacuum break	EARLY	Vacuum broken while the pot is above 150 °C	- HV-110 opened too early - N₂ short so a vent is opened - VP-101 stopped before cool-down	Air (or N ₂ with air) meets hot ester: fire in the column or vapour line; PSHH-102 trips heating but the liquid stays hot	- Procedure: HV-110 only below 150 °C - PSHH-102 trips heating at > 300 mbar (5 s) - HV-110 closed is an arming permissive (prevents arming only) SIF-02 PSHH-102	4	3	12 High	R03 R02 R31 R33	Instrument, Process, Safety, Operations
N8.12	Vacuum break	BEFORE	VP-101 stopped before the vacuum is broken with N ₂	Shutdown steps out of order	Air and pump oil drawn back through VP-101 into the hot column	Shutdown procedure	4	2	8 Medium	R05 R03	Mechanical, Instrument, Process
N8.13	Discharge	EARLY	Residue drained while hot	- Turn-around pressure - residue thought cool	Ester at > 150 °C spills: burns, fire	- Procedure - PPE	3	3	9 Medium	R29 R26	Operations, Safety, Process
N8.14	Discharge	PART OF	Residue not fully drained	- Drain line plugged with solid residue (m.p. 30–38 °C) - hurried turn-around	Degraded residue carried to the next batch: foam, D-102 fouling	Pot level before charging	2	3	6 Medium	R17	Operations
N8.15	Maintenance	OTHER THAN	Column opened with deposits on the hot packing	- Packing not cooled and purged - pyrophoric or polymer deposits	Packing fire when air enters	Permit to work (general)	4	2	8 Medium	R30	Safety, Operations
N8.16	Maintenance	AFTER	Restart after maintenance without a leak test	Leak test omitted after flanges were opened	Air in-leakage on the next heat-up (as N7)	PIC-102 and PV-102 indication	4	2	8 Medium	R01	Operations, Process

Appendix 2 · Drawings

A2.1 Node map of the study (from the HAZOP tab).



A2.2 Screen P&ID; of the OTS, DWG VBD-V1-PID-001 rev A (element capture of the P&ID; tab at 2×). No EPC P&ID; was drawn for this study; the screen P&ID; is the reference drawing. Some tags used in the worksheets are named in the simulator model but shown on the screen P&ID; as unlabelled symbols or inside a loop bubble: TV-101, PV-102, XV-107, FV-104, FV-105, HV-105, the transmitters LT-101, PT-102, FT-108, TT-107 and the trip sensors TE-101B, TE-107B.



Appendix 3 · Action close-out sheets

Action R01 · highest risk addressed 12 High

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R01	N3 N5 N7 N8	Operations / Process	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N3.08	Composition / AS WELL AS: Air enters the hot column (in-leakage)	Flange, thermowell or sight-glass seal leak (air in-leakage up to 12× normal); sample valve opened under vacuum	Hot ester (≈ 255–270 °C) and air at sub-atmospheric pressure: autoignition or fire in the packing or the vapour line	PIC-102 H alarm 200 mbar (PV-102 first closes and hides a small leak); C-101 AIR alarm on the DCS; PSHH-102 at > 300 mbar only for a large leak	12 High
N5.07	Containment / OTHER THAN: Hot ester leak at the P-104 seal	Seal wear; dry running (above); under vacuum the seal also draws air in	Ester at ≈ 180–230 °C sprayed: burns, fire; air drawn into the hot system	Operator rounds; HS-100	6 Medium
N7.03	Composition / AS WELL AS: Air in-leakage into the hot system	Flange or seal leak (up to 12× normal in-leakage); thermowell or sight-glass fitting; P-104 seal under vacuum	Air meets hot ester (≈ 255–270 °C) in the column and vapour line: autoignition or fire; pressure only partly rises and the trip is not reached	PIC-102 H alarm 200 mbar; C-101 AIR alarm on the DCS; PSHH-102 at > 300 mbar (not reached by a 12× leak)	12 High
N8.16	Maintenance / AFTER: Restart after maintenance without a leak test	Leak test omitted after flanges were opened	Air in-leakage on the next heat-up (as N7)	PIC-102 and PV-102 indication	8 Medium

Recommendation

Leak-test C-101 and the vacuum line before every heat-up (pressure-rise test with VP-101 isolated) and after any flange is opened; record the rise rate against an acceptance limit.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R02 · highest risk addressed 12 High

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R02	N3 N6 N7 N8	Instrument / Safety	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N3.08	Composition / AS WELL AS: Air enters the hot column (in-leakage)	Flange, thermowell or sight-glass seal leak (air in-leakage up to 12× normal); sample valve opened under vacuum	Hot ester (≈ 255–270 °C) and air at sub-atmospheric pressure: autoignition or fire in the packing or the vapour line	PIC-102 H alarm 200 mbar (PV-102 first closes and hides a small leak); C-101 AIR alarm on the DCS; PSHH-102 at > 300 mbar only for a large leak	12 High
N6.09	Containment / AS WELL AS: Air drawn in through the D-105 sample valve	Sample valve opened while D-105 is under vacuum	Air into hot condensate and the column (as N3 / N7)	Sampling procedure (isolate before opening)	8 Medium
N7.03	Composition / AS WELL AS: Air in-leakage into the hot system	Flange or seal leak (up to 12× normal in-leakage); thermowell or sight-glass fitting; P-104 seal under vacuum	Air meets hot ester (≈ 255–270 °C) in the column and vapour line: autoignition or fire; pressure only partly rises and the trip is not reached	PIC-102 H alarm 200 mbar; C-101 AIR alarm on the DCS; PSHH-102 at > 300 mbar (not reached by a 12× leak)	12 High
N7.09	Composition / OTHER THAN: Air instead of N ₂ at PV-102 or HV-110	N ₂ cylinder empty and swapped for air; wrong hose connected	The bleed or the vacuum break feeds air into the hot column (as the air in-leakage row)	N ₂ supply procedure	8 Medium
N8.11	Vacuum break / EARLY: Vacuum broken while the pot is above 150 °C	HV-110 opened too early; N ₂ short so a vent is opened; VP-101 stopped before cool-down	Air (or N ₂ with air) meets hot ester: fire in the column or vapour line; PSHH-102 trips heating but the liquid stays hot	Procedure: HV-110 only below 150 °C; PSHH-102 trips heating at > 300 mbar (5 s); HV-110 closed is an arming permissive (prevents arming only)	12 High

Recommendation

Add an O₂ analyser or a leak-rate estimate (from the PV-102 output) on the vacuum line, alarmed. Use LOPA to decide whether an air-ingress SIF (trip heating and purge with N₂) is needed: a 12× leak does not reach PSHH-102.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
Sign-off	Name	Signature	Date	
Action party				
HAZOP leader				
Project lead				

Action R03 · highest risk addressed 12 High

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R03	N7 N8	Instrument / Process	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N7.09	Composition / OTHER THAN: Air instead of N ₂ at PV-102 or HV-110	N ₂ cylinder empty and swapped for air; wrong hose connected	The bleed or the vacuum break feeds air into the hot column (as the air in-leakage row)	N ₂ supply procedure	8 Medium
N8.10	End / cool-down / LESS: Cool-down too slow	HV-104 not opened; P-101 stopped too early (no circulation through D-102)	Long wait under vacuum; pressure to break vacuum early	Shutdown procedure (cool through D-102 and E-104)	6 Medium
N8.11	Vacuum break / EARLY: Vacuum broken while the pot is above 150 °C	HV-110 opened too early; N ₂ short so a vent is opened; VP-101 stopped before cool-down	Air (or N ₂ with air) meets hot ester: fire in the column or vapour line; PSHH-102 trips heating but the liquid stays hot	Procedure: HV-110 only below 150 °C; PSHH-102 trips heating at > 300 mbar (5 s); HV-110 closed is an arming permissive (prevents arming only)	12 High
N8.12	Vacuum break / BEFORE: VP-101 stopped before the vacuum is broken with N ₂	Shutdown steps out of order	Air and pump oil drawn back through VP-101 into the hot column	Shutdown procedure	8 Medium

Recommendation

Interlock HV-110: open only when the pot (TE-101B) is below 150 °C, N₂ supply only (dedicated coupling, no air connection), N₂ low-pressure alarm.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R04 · highest risk addressed 12 High

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R04	N3 N7	Control	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N3.08	Composition / AS WELL AS: Air enters the hot column (in-leakage)	Flange, thermowell or sight-glass seal leak (air in-leakage up to 12× normal); sample valve opened under vacuum	Hot ester ($\approx 255\text{--}270\text{ }^{\circ}\text{C}$) and air at sub-atmospheric pressure: autoignition or fire in the packing or the vapour line	PIC-102 H alarm 200 mbar (PV-102 first closes and hides a small leak); C-101 AIR alarm on the DCS; PSHH-102 at > 300 mbar only for a large leak	12 High
N7.02	Pressure / MORE: Slow vacuum loss: VP-101 capacity falls	Worn vanes; pump oil diluted by condensed ester	PV-102 closes to compensate, then pressure creeps up; hotter pot; less spare capacity to hide a leak	PIC-102 H alarm 200 mbar; PV-102 output on the DCS	6 Medium
N7.03	Composition / AS WELL AS: Air in-leakage into the hot system	Flange or seal leak (up to 12× normal in-leakage); thermowell or sight-glass fitting; P-104 seal under vacuum	Air meets hot ester ($\approx 255\text{--}270\text{ }^{\circ}\text{C}$) in the column and vapour line: autoignition or fire; pressure only partly rises and the trip is not reached	PIC-102 H alarm 200 mbar; C-101 AIR alarm on the DCS; PSHH-102 at > 300 mbar (not reached by a 12× leak)	12 High

Recommendation

Alarm the PV-102 output when it falls below $\approx 20\%$ (assumed): little spare vacuum capacity left, an early sign of a leak or of VP-101 wear.

Response by the action party

Status ☐ Accepted ☐ Accepted with changes ☐ Rejected (give reason) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R05 · highest risk addressed 9 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R05	N4 N7 N8	Mechanical / Instrument	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N4.01	Flow / NO: No tempered water to E-103	P-108 trips (motor, power); HV-105 closed by mistake; loop isolation valve shut	Vapour no longer condensed: ester vapour to VP-101 and the vent, top pressure rises; no reflux	FI-108 L alarm 0.6 m³/h; FSLL-108 trips heating at < 0.36 m³/h (10 s, shares FT-108 with FI-108); PSHH-102 at > 300 mbar	9 Medium
N7.01	Pressure / MORE: Vacuum lost: VP-101 trips	Motor or power failure; overload from ester condensed in the pump oil	Top pressure rises; boiling temperatures rise; PSHH-102 trips heating; after the trip the pressure keeps rising while the column is still hot	PIC-102 H alarm 200 mbar; PSHH-102 trips heating at > 300 mbar (5 s, shares PT-102 with PIC-102)	6 Medium
N7.08	Flow / REVERSE: Back-flow from VP-101 into the column when the pump stops	VP-101 stops without a non-return or isolation valve on its suction	Pump oil mist and air drawn back into the hot column	None on the screen P&ID	6 Medium
N8.12	Vacuum break / BEFORE: VP-101 stopped before the vacuum is broken with N ₂	Shutdown steps out of order	Air and pump oil drawn back through VP-101 into the hot column	Shutdown procedure	8 Medium

Recommendation

Fit a non-return and an automatic isolation valve on the VP-101 suction; add a VP-101 run-status alarm.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R06 · highest risk addressed 9 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R06	N1 N4 N5 N7	Process	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N1.06	Composition / OTHER THAN: Wrong or wet feed charged (water, methanol traces, other FAME)	Biodiesel not dried or methanol not stripped upstream; charge taken from the wrong drum; Table 9 feed charged when the design feed was planned	Water or methanol flash at heat-up: pressure surge, foam, flammable vapour to the VP-101 exhaust; with a different feed the cut points move and fixed cut times give off-spec product	PIC-102 H alarm 200 mbar; slow heat-up procedure; lab sample of the charge (not formalised)	6 Medium
N4.01	Flow / NO: No tempered water to E-103	P-108 trips (motor, power); HV-105 closed by mistake; loop isolation valve shut	Vapour no longer condensed: ester vapour to VP-101 and the vent, top pressure rises; no reflux	FI-108 L alarm 0.6 m³/h; FSLL-108 trips heating at < 0.36 m³/h (10 s, shares FT-108 with FI-108); PSHH-102 at > 300 mbar	9 Medium
N4.07	Heat duty / MORE: Vapour load above the condenser capacity	Duty much higher than design (N2/N3); condenser capacity margin 2.5× the design load (assumed)	Uncondensed vapour to VP-101, pressure rises	E-103 OVLD alarm (condenser overloaded) on the DCS; JIC-101 H alarm 5 kW	6 Medium
N5.01	Level / MORE: D-104 fills up	P-104 trips; LIC-104 left in MAN at a low output; LT-104 reads low; FV-104 stuck closed	Drum full: condensate backs up into E-103, condenser area lost, ester vapour to VP-101 and the trap; top pressure rises; no reflux, packing dries; no trip acts	LIC-104 H alarm 85 %; FIC-104 L alarm 20 kg/h; PIC-102 H alarm 200 mbar; E-103 OVLD alarm	9 Medium
N7.02	Pressure / MORE: Slow vacuum loss: VP-101 capacity falls	Worn vanes; pump oil diluted by condensed ester	PV-102 closes to compensate, then pressure creeps up; hotter pot; less spare capacity to hide a leak	PIC-102 H alarm 200 mbar; PV-102 output on the DCS	6 Medium
N7.06	Flow / MORE: Uncondensed ester to VP-101	Condenser cooling lost (N4); drum full (N5); flooding (N3)	Ester in the pump oil (VP-101 capacity loss); ester mist at the exhaust (flammable vapour indoors); trap fills	Trap ahead of VP-101; E-103 OVLD alarm; FSLL-108 trip for loss of cooling	6 Medium
N7.07	Composition / OTHER THAN: Methanol traces and light esters at the VP-101 exhaust	Unstripped methanol in the charge; lights flashed at heat-up	Flammable vapour indoors near the pump	Trap; general ventilation of the pilot hall	6 Medium

Recommendation

Route the VP-101 exhaust outdoors through a demister; check and empty the trap every batch; consider a cold trap.

Response by the action party

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Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R07 · highest risk addressed 6 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R07	N7	Process / Instrument	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N7.04	Pressure / LESS: Top pressure too low	PIC-102 in MAN with PV-102 closed; PT-102 reads high; N ₂ bleed isolated	Vapour velocity rises at low pressure: column floods, ΔP very high, product off-spec	PIC-102 L alarm 140 mbar; PDI-103 H alarm 18 mbar	6 Medium
N7.05	Flow / MORE: N ₂ bleed too high: PV-102 fails open	Instrument-air failure (PV-102 fails open); PIC-102 in MAN at a high output	Top pressure settles near the PSHH-102 setting: hot pot, slower separation; trip may or may not come (spurious trips)	PIC-102 H alarm 200 mbar; PSHH-102 at > 300 mbar	6 Medium

Recommendation

Review PV-102: when it fails open the top settles at about the PSHH-102 setting. Limit PV-102 travel or move the trip setting so that the outcome is defined.

Response by the action party

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Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R08 · highest risk addressed 8 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R08	N2	Instrument	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N2.04	Temperature / MORE: Hot oil above 290 °C	TT-107 reads low (loose in its thermowell) so TIC-107 drives HE-101 up; TIC-107 in MAN at a high output; heater contactor welded	Oil degradation (coke, acids); oil above its flash point (assumed) so any leak can ignite; hotter D-102 film, pot degradation	TIC-107 H alarm 315 °C (uses TT-107, so blind when TT-107 reads low); TSHH-107 trips heating at > 330 °C on its own sensor TE-107B (5 s); TI-108 return temperature	8 Medium

Recommendation

Add a TT-107 vs TE-107B deviation alarm (e.g. 10 °C, assumed) so that a drifting TT-107 is seen before TSHH-107 trips.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R09 · highest risk addressed 8 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R09	N1 N2	Process / Mechanical	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N1.10	Flow / REVERSE: Hot oil leaks into the process side (D-102 tube leak)	Tube or coil leak in D-102 (oil side at a higher pressure than the pot under vacuum)	Hot oil contaminates residue and lights; cracked oil vapour to the vacuum system; hot-oil inventory falls (N2)	None direct; residue colour / lab sample	4 Low
N2.01	Flow / NO: No hot-oil flow to D-102	P-101 trips (motor, coupling); XV-101 closed (trip or spurious); TV-101 fails closed (air loss); as-built TIC-101 stops P-101	Boil-up stops (safe direction); batch time lost; the heater must not run without flow or the oil in HE-101 overheats locally	HE-101 is enabled only with P-101 running (model, to be confirmed on the plant); JIC-101 duty and TI-108 on the DCS	6 Medium
N2.06	Pressure / MORE: Hot-oil loop over-pressured	TV-101 and XV-101 both closed with P-101 running (dead-head); expansion line blocked; water in new oil flashing at first heat-up	Gasket or seal failure: hot oil at 290 °C sprayed (fire, burns)	Expansion tank and relief on the loop (assumed — not shown on the screen P&ID); TV-101 closes only on trip or by the operator	8 Medium
N2.07	Containment / OTHER THAN: Hot-oil leak at a flange or the P-101 seal	Gasket creep from thermal cycling every batch; P-101 seal wear; flange bolts not re-torqued after maintenance	Hot oil at 290 °C sprays or pools: oil fire, severe burns	HS-100 emergency stop (stops heating, XV-101 closes); insulation; inspection by operators	8 Medium
N2.08	Flow / REVERSE: Cooling water enters the hot oil through an E-104 tube leak	E-104 tube leak while the cooling side pressure is higher than the oil	Water flashes in 290 °C oil: pressure surge in the loop, oil foams, leaks	HV-104 kept closed during the run (cool-down only)	4 Low
N2.09	Composition / AS WELL AS: Air or water in the hot-oil loop	Loop not purged after maintenance; open expansion tank; wet new oil	Oil oxidation and sludge, reduced heat transfer; water flashing at heat-up	Start-up procedure (slow heat-up of the oil)	4 Low

Recommendation

Confirm the hot-oil loop protection: relief and expansion tank, P-101 minimum-flow bypass, hard-wired HE-101 no-flow interlock (flow switch), expansion-tank low-level alarm, oil analysis each year.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R10 · highest risk addressed 8 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R10	N1 N2	Safety / Mechanical	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N1.09	Containment / OTHER THAN: Hot ester released at the pot drain or flange	Drain valve opened or passing; gasket failure after thermal cycling; vacuum broken so the pot is no longer below atmospheric	Ester at $\approx 255\text{--}270\text{ }^{\circ}\text{C}$ (above its flash point, assumed) spills: burns, pool fire	Operation under vacuum (leaks go inward while under vacuum); drain valve closed by procedure	8 Medium
N2.07	Containment / OTHER THAN: Hot-oil leak at a flange or the P-101 seal	Gasket creep from thermal cycling every batch; P-101 seal wear; flange bolts not re-torqued after maintenance	Hot oil at $290\text{ }^{\circ}\text{C}$ sprays or pools: oil fire, severe burns	HS-100 emergency stop (stops heating, XV-101 closes); insulation; inspection by operators	8 Medium

Recommendation

Fit spray shields on hot-oil flanges and the P-101 seal; inspect for leaks before each campaign; extinguisher at HE-101.

Response by the action party

Status ☐ Accepted ☐ Accepted with changes ☐ Rejected (give reason) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R11 · highest risk addressed 3 Low

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R11	N2	Control	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N2.05	Temperature / LESS: Hot oil too cold to hold the duty	Half of the HE-101 elements burnt out (up to –60 %); E-104 / HV-104 left open during the run; TIC-107 set point low	TV-101 runs wide open, duty below set point, longer campaign; the operator may raise R or cut time to compensate	TIC-107 indication; JIC-101 output at 100 % visible on the DCS	3 Low

Recommendation

Alarm when HV-104 (E-104 cooler) is open while heating is armed.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R12 · highest risk addressed 8 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R12	N1 N2 N3 N4 N7 N8	Control / Operations	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N1.05	Pressure / MORE: Pot pressure high (PI-103 rises)	Column flooding (ΔP up to the packing limit); top pressure high (N7); foaming	Pot boiling point rises with pressure; pot temperature and degradation rise; less boil-up for the same duty	PDI-103 H alarm 18 mbar; PIC-102 H alarm 200 mbar; PSHH-102 trips heating at > 300 mbar (top)	6 Medium
N1.07	Composition / AS WELL AS: Foam in the pot (surface-active degradation products)	Residue from the previous batch left in the pot; ester degraded by over-heating; duty raised quickly	Flooding index rises by up to 60 %, heavies carried into the packing and product, ΔP high	PDI-103 H alarm 18 mbar; JIC-101 H alarm 5 kW; flooding index on the DCS	6 Medium
N2.02	Flow / MORE: TV-101 wide open: duty far above the 3 kW set point	JIC-101 left in MAN at a high output; heat-up set point (up to 8 kW) not cut before boiling; TV-101 positioner fault	Duty rises to ≈ 12 kW; column floods, ΔP very high, D-104 fills, vapour reaches VP-101; no trip acts	JIC-101 H alarm 5 kW; PDI-103 H alarm 18 mbar; LIC-104 H alarm 85 %	8 Medium
N2.03	Flow / PART OF: Duty arrives in pulses (as-built on/off heating)	As-built TIC-101 on/off of P-101 (± 1 °C band) used instead of JIC-101	Duty swings between zero and the full loop capacity: flooding bursts, ΔP spikes, poor separation	PDI-103 H alarm 18 mbar; JIC-101 H alarm 5 kW	8 Medium
N3.01	Flow / MORE: Boil-up too high: column floods	Duty set point too high or TV-101 wide open (N2); foam (N1); heat-up duty not cut before boiling; top pressure too low (N7)	Flooding index above 100 %, ΔP rises steeply, stage efficiency falls, heavies reach the top: off-spec off-cut or product; liquid carried to E-103	PDI-103 H alarm 18 mbar; JIC-101 H alarm 5 kW; flooding index display; LIC-104 H alarm 85 %	8 Medium
N3.04	Pressure / MORE: Column ΔP high	Flooding; foam; packing fouled with polymer or crushed	Pot pressure = top + ΔP rises; pot temperature and degradation rise; efficiency falls	PDI-103 H alarm 18 mbar	6 Medium
N4.07	Heat duty / MORE: Vapour load above the condenser capacity	Duty much higher than design (N2/N3); condenser capacity margin 2.5× the design load (assumed)	Uncondensed vapour to VP-101, pressure rises	E-103 OVLD alarm (condenser overloaded) on the DCS; JIC-101 H alarm 5 kW	6 Medium
N7.04	Pressure / LESS: Top pressure too low	PIC-102 in MAN with PV-102 closed; PT-102 reads high; N ₂ bleed isolated	Vapour velocity rises at low pressure: column floods, ΔP very high, product off-spec	PIC-102 L alarm 140 mbar; PDI-103 H alarm 18 mbar	6 Medium
+1 more rows					

Recommendation

Clamp the JIC-101 set point at 5 kW once the pot boils and add a PDI-103 high-high override in the BPCS that cuts duty; train the heat-up duty cut.

Response by the action party

Status ☐ Accepted ☐ Accepted with changes ☐ Rejected (give reason) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R13 · highest risk addressed 9 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R13	N1	Instrument / Safety	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N1.02	Level / LESS: Pot boiled down at the end of the campaign	Product cut not stopped at the end point; richer feed (Table 9 feed, C16:0 58.6 %) distils more; LT-101 reads high	D-102 surface exposed; pot overheats; ester degradation and polymer on D-102 (fouling for later batches); TSHH-101 demand	LI-101 L alarm 30 %; LSLL-101 trips heating at < 20 % (5 s, shares LT-101 with LI-101); TSHH-101 (TE-101B) at > 290 °C; end-of-campaign rule (cumulative TK-202 purity)	9 Medium

Recommendation

Give LSLL-101 its own level transmitter (it shares LT-101 with LI-101); confirm the SIL by LOPA.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R14 · highest risk addressed 9 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R14	N1 N3 N6 N8	Operations / Process	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N1.02	Level / LESS: Pot boiled down at the end of the campaign	Product cut not stopped at the end point; richer feed (Table 9 feed, C16:0 58.6 %) distils more; LT-101 reads high	D-102 surface exposed; pot overheats; ester degradation and polymer on D-102 (fouling for later batches); TSHH-101 demand	LI-101 L alarm 30 %; LSLL-101 trips heating at < 20 % (5 s, shares LT-101 with LI-101); TSHH-101 (TE-101B) at > 290 °C; end-of-campaign rule (cumulative TK-202 purity)	9 Medium
N1.03	Temperature / MORE: Pot temperature high (above the TIC-101 alarm)	Heavies concentrate late in the cut; top pressure high (vacuum loss, N7); hot oil above 290 °C (N2); duty set too high	Thermal degradation of the esters above ≈ 280 °C (assumed limit), dark residue, polymer on D-102, TSHH-101 trip and lost batch time	TIC-101 H alarm 280 °C; TSHH-101 trips heating at > 290 °C on its own sensor TE-101B (5 s); hot oil held at 290 °C by TIC-107 (pot cannot exceed the oil temperature)	9 Medium
N3.05	Temperature / MORE: Heavies rise in the column (top temperature high)	Reflux ratio too low (wrong entry in RY-104); duty high; C18 front reaches the top late in the product cut; packing maldistribution	C18 in TK-202: composite purity below 99 %, whole receiver off-spec	TI-104 and TI-103 trends; lab sample at D-105; end-of-campaign rule	8 Medium
N6.03	Routing / LATE: Product cut stopped too late	End point missed (night shift, lab delay, operator away); TI-103 rise not noticed	C18 front enters TK-202: cumulative purity falls below 99 %, the whole receiver off-spec	TI-103 / TI-104 trends; lab sample; WI-202	8 Medium
N8.08	Product cut / NO: Nobody watching the end of a 15 h campaign	Long campaign over a night shift; operator away	End point missed: C18 into TK-202, pot boiled down	LI-101 L alarm 30 %; LSLL-101 at < 20 %	8 Medium
N8.09	End / cool-down / LATE: Heating not stopped at the end of the campaign	Distillate closed but TIC-107 and JIC-101 left in AUTO	Pot held hot with a small heavy residue: degradation, polymer on D-102, TSHH-101 demand	TIC-101 H alarm 280 °C; TSHH-101 at > 290 °C (TE-101B, 5 s); LSLL-101 at < 20 %	9 Medium

Recommendation

Write the end-of-campaign rule as a stop on the first of: lab result, TI-103 rise, pot level 35 % (assumed); then stop heating at once. Add a TI-103 rate-of-rise alarm.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R15 · highest risk addressed 6 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R15	N1	Instrument / Operations	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N1.01	Level / MORE: Pot overfilled at charging (above the 110 L charge)	Wrong charge volume entered or measured; charge from a second drum added; LT-101 reads low	Liquid close to the column bottom; residue entrained into the packing at first boil; heavies (C18) in the off-cut and product; foam-over at heat-up	LI-101 indication only (no high alarm); charging procedure 110 L	6 Medium

Recommendation

Add a high-level alarm on LI-101 for charging and charge through a meter or by weight.

Response by the action party

Status ☐ Accepted ☐ Accepted with changes ☐ Rejected (give reason) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R16 · highest risk addressed 6 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R16	N1 N4 N5 N7 N8	Operations / Laboratory	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N1.06	Composition / OTHER THAN: Wrong or wet feed charged (water, methanol traces, other FAME)	Biodiesel not dried or methanol not stripped upstream; charge taken from the wrong drum; Table 9 feed charged when the design feed was planned	Water or methanol flash at heat-up: pressure surge, foam, flammable vapour to the VP-101 exhaust; with a different feed the cut points move and fixed cut times give off-spec product	PIC-102 H alarm 200 mbar; slow heat-up procedure; lab sample of the charge (not formalised)	6 Medium
N4.05	Flow / REVERSE: Water enters the process side (E-103 tube leak)	Tube leak; process side is under vacuum so water leaks inward	Water flashes in E-103/D-104: pressure rises, P-104 cavitates, wet product	PIC-102 H alarm 200 mbar; lab sample of the distillate	4 Low
N5.08	Composition / AS WELL AS: Water or methanol in the drum at start-up	Wet or unstripped charge (N1); E-103 tube leak (N4)	P-104 vapour-locks or cavitates; pressure swings; flammable vapour to VP-101	PIC-102 H alarm 200 mbar	4 Low
N7.07	Composition / OTHER THAN: Methanol traces and light esters at the VP-101 exhaust	Unstripped methanol in the charge; lights flashed at heat-up	Flammable vapour indoors near the pump	Trap; general ventilation of the pilot hall	6 Medium
N8.01	Charging / OTHER THAN: Different feed charged than planned	Table 9 feed (C16:0 58.6 %) or another lot charged; no lab analysis	Cut points move: a cut plan fixed in time gives off-spec product or lost C16:0	Lab sample of D-105 during the cuts	6 Medium

Recommendation

Analyse every charge before charging (water, methanol, C16:0) and adapt the cut plan to the feed.

Response by the action party

Status ☐ Accepted ☐ Accepted with changes ☐ Rejected (give reason) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R17 · highest risk addressed 6 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R17	N1 N3 N8	Operations	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N1.07	Composition / AS WELL AS: Foam in the pot (surface-active degradation products)	Residue from the previous batch left in the pot; ester degraded by over-heating; duty raised quickly	Flooding index rises by up to 60 %, heavies carried into the packing and product, ΔP high	PDI-103 H alarm 18 mbar; JIC-101 H alarm 5 kW; flooding index on the DCS	6 Medium
N3.04	Pressure / MORE: Column ΔP high	Flooding; foam; packing fouled with polymer or crushed	Pot pressure = top + ΔP rises; pot temperature and degradation rise; efficiency falls	PDI-103 H alarm 18 mbar	6 Medium
N8.14	Discharge / PART OF: Residue not fully drained	Drain line plugged with solid residue (m.p. 30–38 °C); hurried turn-around	Degraded residue carried to the next batch: foam, D-102 fouling	Pot level before charging	6 Medium

Recommendation

Drain the residue fully every batch, inspect D-102, and keep an antifoam procedure.

Response by the action party

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Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R18 · highest risk addressed 9 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R18	N1	Process	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N1.03	Temperature / MORE: Pot temperature high (above the TIC-101 alarm)	Heavies concentrate late in the cut; top pressure high (vacuum loss, N7); hot oil above 290 °C (N2); duty set too high	Thermal degradation of the esters above ≈ 280 °C (assumed limit), dark residue, polymer on D-102, TSHH-101 trip and lost batch time	TIC-101 H alarm 280 °C; TSHH-101 trips heating at > 290 °C on its own sensor TE-101B (5 s); hot oil held at 290 °C by TIC-107 (pot cannot exceed the oil temperature)	9 Medium
N1.08	Heat duty / LESS: Reboiler D-102 fouled (polymer on the heating surface)	Pot over-heated above ≈ 285 °C in earlier batches; residue not fully drained; no cleaning interval	UA falls (up to –95 %); TV-101 runs wide open; duty cannot be held; longer campaign; hotter film makes more polymer	JIC-101 duty with TV-101 output; TI-101 (reboiler outlet) minus TI-106 shows the film ΔT	6 Medium

Recommendation

Track the reboiler UA per batch (duty vs TV-101 output and TI-101 – TI-106); clean D-102 when it falls below 70 % of new (assumed).

Response by the action party

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Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R19 · highest risk addressed 9 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R19	N4	Instrument / Process	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N4.03	Temperature / LESS: Tempered water too cold (below 45 °C)	Tempered-water TCV fails (supply falls to ≈ 22 °C); loop heater off; cold water used at start-up	C16/C18 esters freeze on the tubes; E-103 plugs, condenser capacity and vacuum lost, column upset, ester to VP-101	TI-109 L alarm 45 °C; TI-105 L alarm 45 °C; no trip	9 Medium

Recommendation

Add a TI-109 low-low alarm and make the tempered-water TCV fail towards warm; consider stopping heating when TI-109 stays below 40 °C (assumed).

Response by the action party

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Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R20 · highest risk addressed 9 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R20	N4	Mechanical	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N4.02	Flow / LESS: Tempered-water flow low	HV-105 strainer blocking (flow down by up to 80 %); HIC-105 set too low; P-108 worn	Condenser capacity falls; part of the vapour passes to VP-101; FSLL-108 trips heating when the flow falls below its setting	FI-108 L alarm 0.6 m³/h; FSLL-108 at < 0.36 m³/h (10 s)	9 Medium

Recommendation

Fit a duplex strainer with a ΔP gauge upstream of HV-105.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R21 · highest risk addressed 9 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R21	N4	Operations	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N4.03	Temperature / LESS: Tempered water too cold (below 45 °C)	Tempered-water TCV fails (supply falls to ≈ 22 °C); loop heater off; cold water used at start-up	C16/C18 esters freeze on the tubes: E-103 plugs, condenser capacity and vacuum lost, column upset, ester to VP-101	TI-109 L alarm 45 °C; TI-105 L alarm 45 °C; no trip	9 Medium
N4.06	Phase / AS WELL AS: Frozen ester left in E-103 from the last batch	Tempered water left running cold after shutdown; cold start with water below 45 °C	Next start-up with a partly plugged condenser: high pressure, vapour to VP-101	TI-109 L alarm 45 °C; start-up procedure	6 Medium

Recommendation

Warm E-103 to 60 °C before the first vapour arrives and never leave the tempered water cold after shutdown.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R22 · highest risk addressed 9 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R22	N3 N5 N7	Instrument / Safety	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N3.03	Flow / NO: No reflux to the packing	P-104 trips; LSLL-104 stops P-104; drum emptied by a high-reading LT-104	Packing dries from the top; top temperature rises; heavies reach the drum; drum and E-103 fill (N5)	FIC-104 L alarm 20 kg/h; LIC-104 L/H alarms 20 / 85 %	6 Medium
N5.01	Level / MORE: D-104 fills up	P-104 trips; LIC-104 left in MAN at a low output; LT-104 reads low; FV-104 stuck closed	Drum full: condensate backs up into E-103, condenser area lost, ester vapour to VP-101 and the trap; top pressure rises; no reflux, packing dries; no trip acts	LIC-104 H alarm 85 %; FIC-104 L alarm 20 kg/h; PIC-102 H alarm 200 mbar; E-103 OVLD alarm	9 Medium
N5.02	Level / LESS: D-104 runs dry	LT-104 drifts high (up to +40 % of span) so LIC-104 draws the drum down; FT-105 reads low (more drawn than shown); duty lost	P-104 runs dry (seal and impeller damage), reflux lost; LSLL-104 stops P-104 and the case then becomes "drum fills"	LIC-104 L alarm 20 %; LSLL-104 stops P-104 at < 5 % (5 s, equipment interlock, not a SIF); P-104 DRY alarm	6 Medium
N7.06	Flow / MORE: Uncondensed ester to VP-101	Condenser cooling lost (N4); drum full (N5); flooding (N3)	Ester in the pump oil (VP-101 capacity loss); ester mist at the exhaust (flammable vapour indoors); trap fills	Trap ahead of VP-101; E-103 OVLD alarm; FSLL-108 trip for loss of cooling	6 Medium

Recommendation

Add LSHH-104 (D-104 high-high) or use the E-103 overload signal to cut heating; LOPA to size it. A full drum sends ester to VP-101 with no trip.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R23 · highest risk addressed 8 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R23	N3 N5 N8	Control / Operations	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N3.05	Temperature / MORE: Heavies rise in the column (top temperature high)	Reflux ratio too low (wrong entry in RY-104); duty high; C18 front reaches the top late in the product cut; packing maldistribution	C18 in TK-202: composite purity below 99 %, whole receiver off-spec	TI-104 and TI-103 trends; lab sample at D-105; end-of-campaign rule	8 Medium
N5.03	Reflux ratio / LESS: Real reflux ratio below the RY-104 value	FT-104 reads high (up to +40 %); FV-104 sticks; R typed as 3 instead of 10 at a shift change	Heavies (product cut) or C16 (off-cut) leave at the top: product off-spec or C16:0 lost to the off-cut	Lab sample at D-105; TI-104 / TI-103 trends; FIC-104 L alarm 20 kg/h	8 Medium
N5.05	Flow / MORE: Distillate rate too high	FT-105 reads low (up to -40 %); FIC-105 in MAN at a high output; R too low	Purity of the cut falls; drum drawn down	LIC-104 L alarm 20 %; WI-201 / WI-202 rate; lab sample	6 Medium
N8.07	Product cut / AS WELL AS: Shift handover during the product cut	Wrong R entered (3 instead of 10); HS-107 position not handed over	Product off-spec or sent to the wrong receiver	Lab sample; WI-202	8 Medium

Recommendation

Limit RY-104 entries (off-cut 10–25, product 8–20, assumed) and add R and HS-107 to the shift handover check.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R24 · highest risk addressed 8 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R24	N5 N6 N8	Instrument	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N5.05	Flow / MORE: Distillate rate too high	FT-105 reads low (up to –40 %); FIC-105 in MAN at a high output; R too low	Purity of the cut falls; drum drawn down	LIC-104 L alarm 20 %; WI-201 / WI-202 rate; lab sample	6 Medium
N6.01	Routing / OTHER THAN: Distillate goes to the wrong receiver	HS-107 set to the wrong position; XV-107 fails to move (HS-107 changes the indication only); receivers swapped after maintenance	Lights in TK-202 (the whole product receiver off-spec) or product lost to TK-201	WI-201 / WI-202 weights (which tank is rising); lab sample	8 Medium
N8.07	Product cut / AS WELL AS: Shift handover during the product cut	Wrong R entered (3 instead of 10); HS-107 position not handed over	Product off-spec or sent to the wrong receiver	Lab sample; WI-202	8 Medium

Recommendation

Fit XV-107 position switches with a discrepancy alarm against HS-107, and a WI-201 / WI-202 rate check.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
Sign-off	Name	Signature	Date	
Action party				
HAZOP leader				
Project lead				

Action R25 · highest risk addressed 8 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R25	N3 N6 N8	Operations / Process	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N3.06	Temperature / LESS: TI-104 reads low (measurement)	TT-104 loose in its thermowell (reads up to 25 °C low)	The top looks "clean" too early: switch to product while lights are still present; lights in TK-202	TI-103 for comparison; lab sample of D-105 before switching	6 Medium
N6.02	Routing / EARLY: Switch to product too early	TI-104 reads low; switch made on time instead of the sample; lab result read from an earlier sample	Light esters in TK-202: composite purity below 99 %	Switch rule (D-104 light esters < 4 %); lab GC of D-105 (≈ 10 min delay)	8 Medium
N6.08	Composition / PART OF: Sample at D-105 not representative	Sample line not flushed (dead leg holds an earlier cut)	Wrong switch or stop decision	Sampling procedure	6 Medium
N6.09	Containment / AS WELL AS: Air drawn in through the D-105 sample valve	Sample valve opened while D-105 is under vacuum	Air into hot condensate and the column (as N3 / N7)	Sampling procedure (isolate before opening)	8 Medium
N8.06	Off-cut / AFTER: Decision taken on a lab result that arrives after the change it should guide	GC result ≈ 10 min after sampling; sample taken late	Switch to product late (C16:0 lost) or early (lights in product)	Switch rule; TI-104 trend	6 Medium

Recommendation

Switch cuts on the D-105 lab result, not on TI-104 alone; flush the sample line; use a closed sample pot isolated from the vacuum.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R26 · highest risk addressed 9 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R26	N5 N6 N8	Safety / Process	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N5.07	Containment / OTHER THAN: Hot ester leak at the P-104 seal	Seal wear; dry running (above); under vacuum the seal also draws air in	Ester at $\approx 180\text{--}230\text{ }^{\circ}\text{C}$ sprayed: burns, fire; air drawn into the hot system	Operator rounds; HS-100	6 Medium
N6.07	Temperature / MORE: Hot distillate at the sample point and receivers	No product cooler: distillate at $\approx \text{TI-105}$ ($227\text{ }^{\circ}\text{C}$ in the product cut) reaches D-105 and the receivers	Burns when sampling or changing receivers; ester vapour at the sample point	PPE and sampling procedure	9 Medium
N8.13	Discharge / EARLY: Residue drained while hot	Turn-around pressure; residue thought cool	Ester at $> 150\text{ }^{\circ}\text{C}$ spills: burns, fire	Procedure; PPE	9 Medium

Recommendation

Add a sample cooler at D-105 and a product cooler before TK-201 / TK-202; face shield and gloves for sampling and draining.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R27 · highest risk addressed 6 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R27	N6	Process / Instrument	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N6.06	Level / MORE: Receiver overfilled	Receiver smaller than the cut (Table 9 feed gives a larger product cut); weight not watched	Hot ester overflows: burns, fire risk	WI-201 / WI-202 weights (no alarm)	6 Medium

Recommendation

Add WI-201 / WI-202 high-weight alarms and size the receivers for the largest cut (Table 9 feed).

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R28 · highest risk addressed 4 Low

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R28	N6	Process / Operations	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N6.05	Flow / REVERSE: Back-flow from a receiver into the column or drum	Receiver vented to atmosphere or N ₂ while the column is under vacuum; XV-107 passing when P-104 stops	Off-cut or product sucked back; N ₂ or air drawn into the hot drum and column	Receivers on the vacuum system (assumed); XV-107 CLOSED at total reflux	4 Low

Recommendation

Keep the receivers balanced to the vacuum system with a non-return valve; put XV-107 to CLOSED before any receiver is changed.

Response by the action party

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Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R29 · highest risk addressed 9 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R29	N1 N8	Operations	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N1.09	Containment / OTHER THAN: Hot ester released at the pot drain or flange	Drain valve opened or passing; gasket failure after thermal cycling; vacuum broken so the pot is no longer below atmospheric	Ester at $\approx 255\text{--}270\text{ }^{\circ}\text{C}$ (above its flash point, assumed) spills: burns, pool fire	Operation under vacuum (leaks go inward while under vacuum); drain valve closed by procedure	8 Medium
N8.02	Charging / BEFORE: Charging into a pot still hot from the last batch	Turn-around shortened; residue not cooled	Fresh ester and air on hot surfaces: vapour release, fire	Cool-down procedure	8 Medium
N8.13	Discharge / EARLY: Residue drained while hot	Turn-around pressure; residue thought cool	Ester at $> 150\text{ }^{\circ}\text{C}$ spills: burns, fire	Procedure; PPE	9 Medium

Recommendation

Drain the residue only below $80\text{ }^{\circ}\text{C}$ (assumed) through a closed drain; double-block the pot drain; do not charge a hot pot.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R30 · highest risk addressed 8 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R30	N3 N8	Safety / Operations	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N3.07	Composition / PART OF: Fewer effective stages than designed	Packing maldistribution or damage (efficiency down by up to 60 %); liquid distributor partly plugged with frozen ester or polymer	Purity of the product cut below 99 %; smaller on-spec cut; lower recovery	Temperature profile TI-104 / TI-103 / TI-102; lab results	6 Medium
N8.15	Maintenance / OTHER THAN: Column opened with deposits on the hot packing	Packing not cooled and purged; pyrophoric or polymer deposits	Packing fire when air enters	Permit to work (general)	8 Medium

Recommendation

Before opening C-101: cool below 50 °C, purge with N₂ and keep the packing wet; permit to work.

Response by the action party

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Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Project lead			

Action R31 · highest risk addressed 12 High

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R31	N1 N2 N3 N4 N7 N8	Safety	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N1.02	Level / LESS: Pot boiled down at the end of the campaign	Product cut not stopped at the end point; richer feed (Table 9 feed, C16:0 58.6 %) distils more; LT-101 reads high	D-102 surface exposed; pot overheats; ester degradation and polymer on D-102 (fouling for later batches); TSHH-101 demand	LI-101 L alarm 30 %; LSLL-101 trips heating at < 20 % (5 s, shares LT-101 with LI-101); TSHH-101 (TE-101B) at > 290 °C; end-of-campaign rule (cumulative TK-202 purity)	9 Medium
N1.03	Temperature / MORE: Pot temperature high (above the TIC-101 alarm)	Heavies concentrate late in the cut; top pressure high (vacuum loss, N7); hot oil above 290 °C (N2); duty set too high	Thermal degradation of the esters above ≈ 280 °C (assumed limit), dark residue, polymer on D-102, TSHH-101 trip and lost batch time	TIC-101 H alarm 280 °C; TSHH-101 trips heating at > 290 °C on its own sensor TE-101B (5 s); hot oil held at 290 °C by TIC-107 (pot cannot exceed the oil temperature)	9 Medium
N2.04	Temperature / MORE: Hot oil above 290 °C	TT-107 reads low (loose in its thermowell) so TIC-107 drives HE-101 up; TIC-107 in MAN at a high output; heater contactor welded	Oil degradation (coke, acids); oil above its flash point (assumed) so any leak can ignite; hotter D-102 film, pot degradation	TIC-107 H alarm 315 °C (uses TT-107, so blind when TT-107 reads low); TSHH-107 trips heating at > 330 °C on its own sensor TE-107B (5 s); TI-108 return temperature	8 Medium
N3.08	Composition / AS WELL AS: Air enters the hot column (in-leakage)	Flange, thermowell or sight-glass seal leak (air in-leakage up to 12× normal); sample valve opened under vacuum	Hot ester (≈ 255–270 °C) and air at sub-atmospheric pressure: autoignition or fire in the packing or the vapour line	PIC-102 H alarm 200 mbar (PV-102 first closes and hides a small leak); C-101 AIR alarm on the DCS; PSHH-102 at > 300 mbar only for a large leak	12 High
N4.01	Flow / NO: No tempered water to E-103	P-108 trips (motor, power); HV-105 closed by mistake; loop isolation valve shut	Vapour no longer condensed: ester vapour to VP-101 and the vent, top pressure rises; no reflux	FI-108 L alarm 0.6 m³/h; FSLL-108 trips heating at < 0.36 m³/h (10 s, shares FT-108 with FI-108); PSHH-102 at > 300 mbar	9 Medium
N4.02	Flow / LESS: Tempered-water flow low	HV-105 strainer blocking (flow down by up to 80 %); HIC-105 set too low; P-108 worn	Condenser capacity falls; part of the vapour passes to VP-101; FSLL-108 trips heating when the flow falls below its setting	FI-108 L alarm 0.6 m³/h; FSLL-108 at < 0.36 m³/h (10 s)	9 Medium
N7.03	Composition / AS WELL AS: Air in-leakage into the hot system	Flange or seal leak (up to 12× normal in-leakage); thermowell or sight-glass fitting; P-104 seal under vacuum	Air meets hot ester (≈ 255–270 °C) in the column and vapour line: autoignition or fire; pressure only partly rises and the trip is not reached	PIC-102 H alarm 200 mbar; C-101 AIR alarm on the DCS; PSHH-102 at > 300 mbar (not reached by a 12× leak)	12 High
N7.06	Flow / MORE: Uncondensed ester to VP-101	Condenser cooling lost (N4); drum full (N5); flooding (N3)	Ester in the pump oil (VP-101 capacity loss); ester mist at the exhaust (flammable vapour indoors); trap fills	Trap ahead of VP-101; E-103 OVLD alarm; FSLL-108 trip for loss of cooling	6 Medium
+2 more rows					

Recommendation

Carry every High row and every row that relies on a SIS-100 trip into LOPA; record the SIL targets in the SRS.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
Sign-off				
Action party				
HAZOP leader				
Project lead				

Action R32 · highest risk addressed 8 Medium

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R32	N6 N8	Operations	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N6.02	Routing / EARLY: Switch to product too early	TI-104 reads low; switch made on time instead of the sample; lab result read from an earlier sample	Light esters in TK-202: composite purity below 99 %	Switch rule (D-104 light esters < 4 %); lab GC of D-105 (≈ 10 min delay)	8 Medium
N6.03	Routing / LATE: Product cut stopped too late	End point missed (night shift, lab delay, operator away); TI-103 rise not noticed	C18 front enters TK-202: cumulative purity falls below 99 %, the whole receiver off-spec	TI-103 / TI-104 trends; lab sample; WI-202	8 Medium
N8.08	Product cut / NO: Nobody watching the end of a 15 h campaign	Long campaign over a night shift; operator away	End point missed: C18 into TK-202, pot boiled down	LI-101 L alarm 30 %; LSL-101 at < 20 %	8 Medium

Recommendation

Alarm the operator at the cut switch and the end point (lab request reminders); staff night shifts of long campaigns.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
Sign-off	Name	Signature	Date	
Action party				
HAZOP leader				
Project lead				

Action R33 · highest risk addressed 12 High

Phase	Action no.	Node	Action party	Drawings
Design / teaching pilot	R33	N1 N2 N3 N4 N5 N7 N8	Operations / Safety	VBD-V1-PID-001 rev A; node map (App. 2)

Raised from	Deviation	Causes	Consequences	Safeguards	Risk
N1.03	Temperature / MORE: Pot temperature high (above the TIC-101 alarm)	Heavies concentrate late in the cut; top pressure high (vacuum loss, N7); hot oil above 290 °C (N2); duty set too high	Thermal degradation of the esters above ≈ 280 °C (assumed limit), dark residue, polymer on D-102, TSHH-101 trip and lost batch time	TIC-101 H alarm 280 °C; TSHH-101 trips heating at > 290 °C on its own sensor TE-101B (5 s); hot oil held at 290 °C by TIC-107 (pot cannot exceed the oil temperature)	9 Medium
N1.05	Pressure / MORE: Pot pressure high (PI-103 rises)	Column flooding (ΔP up to the packing limit); top pressure high (N7); foaming	Pot boiling point rises with pressure; pot temperature and degradation rise; less boil-up for the same duty	PDI-103 H alarm 18 mbar; PIC-102 H alarm 200 mbar; PSHH-102 trips heating at > 300 mbar (top)	6 Medium
N1.07	Composition / AS WELL AS: Foam in the pot (surface-active degradation products)	Residue from the previous batch left in the pot; ester degraded by over-heating; duty raised quickly	Flooding index rises by up to 60 %, heavies carried into the packing and product, ΔP high	PDI-103 H alarm 18 mbar; JIC-101 H alarm 5 kW; flooding index on the DCS	6 Medium
N2.01	Flow / NO: No hot-oil flow to D-102	P-101 trips (motor, coupling); XV-101 closed (trip or spurious); TV-101 fails closed (air loss); as-built TIC-101 stops P-101	Boil-up stops (safe direction); batch time lost; the heater must not run without flow or the oil in HE-101 overheats locally	HE-101 is enabled only with P-101 running (model, to be confirmed on the plant); JIC-101 duty and TI-108 on the DCS	6 Medium
N2.02	Flow / MORE: TV-101 wide open: duty far above the 3 kW set point	JIC-101 left in MAN at a high output; heat-up set point (up to 8 kW) not cut before boiling; TV-101 positioner fault	Duty rises to ≈ 12 kW; column floods, ΔP very high, D-104 fills, vapour reaches VP-101; no trip acts	JIC-101 H alarm 5 kW; PDI-103 H alarm 18 mbar; LIC-104 H alarm 85 %	8 Medium
N2.04	Temperature / MORE: Hot oil above 290 °C	TT-107 reads low (loose in its thermowell) so TIC-107 drives HE-101 up; TIC-107 in MAN at a high output; heater contactor welded	Oil degradation (coke, acids); oil above its flash point (assumed) so any leak can ignite; hotter D-102 film, pot degradation	TIC-107 H alarm 315 °C (uses TT-107, so blind when TT-107 reads low); TSHH-107 trips heating at > 330 °C on its own sensor TE-107B (5 s); TI-108 return temperature	8 Medium
N3.01	Flow / MORE: Boil-up too high: column floods	Duty set point too high or TV-101 wide open (N2); foam (N1); heat-up duty not cut before boiling; top pressure too low (N7)	Flooding index above 100 %, ΔP rises steeply, stage efficiency falls, heavies reach the top: off-spec off-cut or product; liquid carried to E-103	PDI-103 H alarm 18 mbar; JIC-101 H alarm 5 kW; flooding index display; LIC-104 H alarm 85 %	8 Medium
N3.03	Flow / NO: No reflux to the packing	P-104 trips; LSLL-104 stops P-104; drum emptied by a high-reading LT-104	Packing dries from the top; top temperature rises; heavies reach the drum; drum and E-103 fill (N5)	FIC-104 L alarm 20 kg/h; LIC-104 L/H alarms 20 / 85 %	6 Medium
+12 more rows					

Recommendation

Train the deviations of the dynamic check on the simulator (one per shift team) and revalidate this HAZOP every 5 years or at any MOC.

Response by the action party

Status	☐ Accepted	☐ Accepted with changes	☐ Rejected (give reason)	☐ Closed
Sign-off	Name	Signature	Date	
Action party				
HAZOP leader				
Project lead				